Season 3 — Four Ways To Fail

Research document. Nineteen episodes. The instrument from Season 2 meets the texts, four times, and loses three of them. This is where the maths gets taught, because every failure has an exact mechanism and the mechanism is the lesson.

> *Two anchors per concept, as in Season 2.* ***Character*** *for the audience,* ***clinical*** *for you. And from here every episode carries the number that caused the failure, not a description of it.*

The shape of the season

Four rounds. Each one asks a sharper question than the last, and each one fails for a reason that is worth more than the answer would have been.

ROUND	THE QUESTION IT ASKED	RESULT	THE REASON
1	do the four stereotypes appear together in persecution myths	**FAIL**	the metric collapses on short texts
2	again, on 71 texts never scored, metric fixed in advance	**4 of 6**	one clean number, two informative failures
3	do the four **converge on one passage**	**FAIL, 1 of 5**	word order turns out to be irrelevant to the statistic
4	is there a **dissenting voice**, the thing Girard actually claims	**FAIL**	the round's own sensitivity analyses destroy it

Which episodes are load-bearing

Episode 12. What a null holds constant. It is the single most transferable idea in the entire series and everything after it depends on having it.

Episode 10. The one surviving positive result, and what it does not mean. If the audience takes only one number away it will be this one, so the limits have to be stated in the same breath.

Episode 8. The noise floor. One subtraction, no new data, and it disarms a confound the project carried for three seasons. It is the cheapest transferable move in the series: **read your controls as rulers, not as gates.**

Episode 19. Two failed rounds, one cause. The whole season resolves into one sentence about rates and position.

What is not here

No conclusions about Girard. Three rounds failed and one half-passed, and what that means about the theory is genuinely open at the end of this season. Season 4 runs three more rounds before anything can be said.

Collecting Season 2

Thirteen episodes built an instrument and tested nothing. Written blind, frozen, corpus audited, tokens fixed, lexicon de-noised, metric chosen for the right reason and validated against a false positive it caught by itself.

Every decision was defensible. Most of them were right.

This episode is what happened when it met the texts.

The setup

Ovid's *Metamorphoses*, cut into 31 individual myths. Seventeen sorted into tiers **before anything was scored**, on Girardian grounds only.

Notice what that sorting commits you to. **It is a public list of which texts you expect to win.** Not an interpretation offered after the fact, not a reading, but a partition of the material written down while the answer is still unknown, by somebody who then has to live with it.

HIGH, where the mechanism should be: Pentheus torn apart by his mother. Actaeon torn apart by his hounds. Lycaon, cannibalism and expulsion. Cadmus, where sown men slaughter each other.

LOW, the controls: Narcissus and Phaethon, **both declared controls in the fetching script before the corpus existed**, plus the creation, the four ages, Daphne, Ceres, Aglauros.

And the falsification conditions, written down first:

> Any LOW text scoring above any HIGH text. HIGH and LOW overlapping so much no threshold separates them. The null test matching the buckets' separation.

Read those as what they are, which is a list of ways to lose that the author wrote for himself. **Season 2 episode 1 said the whole season was an attempt to build a test that can come back negative.** This is the moment the test acquires the ability, and the three conditions are the exact shape of the negative it is now capable of returning.

The result

TIER	N	MEAN MIN
HIGH	3	0.22
MID	6	0.19
LOW	6	**0.25**

No separation. And the controls are highest.

Take the second half of that sentence slowly, because it is worse than a null result. A null result is HIGH and LOW landing on top of each other, which says the instrument cannot tell them apart. **This says the instrument can tell them apart and has them the wrong way round.** The texts chosen because nothing Girardian happens in them outscored the texts chosen because the mechanism is at its most naked.

Two falsification conditions fired at once:

- **Aglauros**, a declared control, scores **MIN 1.27**, beating **every** HIGH text. She is a minor Athenian figure in Ovid who is turned to stone out of envy. **No crowd, no accusation, nobody killed by anybody**, which is exactly why it was put in the LOW pile before scoring.
- **Pentheus**, Girard's single most important persecution myth, scores **MIN 0.00.**

And 0.00 is not a low score. **It is the instrument reporting that at least one stereotype is entirely absent** from the text Girard uses to explain what all four look like when they occur together.

The device: two scoreboards, only one of which was written down

Put them side by side on screen.

	HIGH	MID	LOW	VERDICT
MIN, the registered metric	0.22	0.19	**0.25**	**FAIL**
SUM, not registered	**9.08**	6.06	**3.52**	perfect

> **Look at the second row.** HIGH above MID above LOW, in exactly the predicted order, cleanly separated, no overlap. **It is a beautiful result.**
>
> **And it is not the metric that was written down.**

Both rows come from the same scoring run on the same texts with the same lexicon. **Nothing separates them except which of two arithmetic operations was performed on four numbers**, and one of those operations was named in a file committed before any text was scored.

What happens next is the episode

Everything in Season 2 was building toward a moment nobody plans for: **you have a failed prediction and a beautiful unregistered result sitting next to it.**

The temptation is not to lie. Nobody in this story is contemplating a lie. **The temptation is to notice that SUM is obviously the better metric**, which it might genuinely be, and to lead with it, mentioning MIN somewhere lower down in a paragraph about robustness.

And every sentence you would write while doing that is true. SUM does order the tiers. The ordering is clean. MIN does have a known weakness, which the next episode is about, and it is a real weakness discovered for real reasons. **You would not have to write down a single false statement to publish this as a success.**

> From the results file: **"Switching metrics after seeing the result is precisely the move pre-registration exists to prevent."**

The sum was not claimed. That is episode 1's real content, and it costs the project its first result.

The clinical anchor

A trial whose primary endpoint fails and whose secondary endpoint is beautiful.

Everyone knows what happens next in the literature. Everyone knows what should happen. **The gap between those two sentences is the reason this series exists.**

And notice the specific thing that goes wrong when a secondary is promoted, because it is not that the secondary is untrue. **It is that nobody can tell how many secondaries there were.** One endpoint out of one is evidence. The same endpoint, selected from twelve after the fact, is close to nothing, and the published paper looks identical either way.

The character anchor

A goalkeeper in a penalty shootout who dives the wrong way and the ball hits him.

The save is completely real. It is in the record, it wins the match, and the crowd's reaction is not misplaced. **What did not happen is the thing he was being tested on**, which was reading the striker, and the only person in the stadium who knows the difference is him.

Now give him a microphone afterwards. **Every honest description of the save is also a description of his judgement**, because the save and the judgement are the same event to everybody watching. He would have to interrupt his own good news to separate them, and nobody in the room is asking him to.

SUM is the ball hitting him. The result is real and the project did not read the play.

What would count against this

1. **MIN being a bad metric.** It might be, and this is now the strongest item on the list. **The argument has to come from the theory rather than from the fact that it failed**, and Season 2 episode 12 made exactly that argument. **It also overstated it.** Girard's own rule, in *The Scapegoat* chapter 3, is "not all the stereotypes must be present: three are enough and often even two," which is **N-of-M and not a minimum.** So MIN is a deliberate strengthening of the source rather than a reading of it, and **the round you are watching fail is failing at a bar the theory never set.** That does not rescue the prediction, because the prediction was registered with MIN in it. **It does mean the failure is a fact about this instrument and not about Girard**, and the series should say so here rather than letting it look like a defeat for the theory.
2. **The tiers being wrong.** If Pentheus is not a persecution myth, the prediction was mis-specified rather than the instrument being blind. **Season 1 episode 15 read the play and found the accusation stereotype genuinely weak in it**, which is a real version of this objection.

Handing off

Episode 2 explains why MIN collapsed, and the explanation is one line of arithmetic with nothing Girardian in it at all.

Questions

1. **The sum ordered the tiers perfectly.** Give the strongest argument for reporting it, then say why it loses.
2. **Aglauros is a control that scored highest.** Guess why, before you are told.

3. **Pentheus at zero.** Is that a problem with Girard, the lexicon, or the metric? Three very different answers.

This is the first maths episode and it teaches one tool from zero.

Collecting episode 1

MIN collapsed. **Not because Girard is wrong, and not because the texts were wrong.** Because of a property of minimums that nobody thought about, and it takes one idea to see it.

The device: raisins in a cake

Do this before any formula.

A baker puts **100 raisins** into a cake and cuts it into **100 slices.**

How many raisins in the average slice? One.

Now the question that matters: **how many slices have NO raisin at all?**

Most people say none, or a handful. **The answer is about 37.**

Ask why the wrong answer feels right, because that is the useful part. An average of one invites you to picture one raisin per slice, tidily. **But nothing is arranging them.** Each raisin lands wherever the batter carries it, with no memory of where the others went, and a slice that already has two is exactly as likely to receive the next one as a slice that has none.

> Raisins do not queue up politely. Some slices get three, some get two, and **more than a third get zero, even though the average is one.**

That is the entire idea. Rare things clump, and the chance of getting none is much higher than the average suggests.

The character anchor

Gacha pulls.

A banner advertises a 1% rate. You save up a hundred pulls, which is exactly the expected number for one copy, and you get nothing, and the comment section fills with people certain the rate is a lie.

It is not a lie. At 1% over a hundred pulls, coming away empty is the ordinary outcome, and it happens to roughly a third of everybody who tries.

And here is the detail worth putting on screen. **Every one of those games ships a pity counter**, a guarantee after so many failures. That mechanism exists because studios discovered that honest arithmetic at the advertised rate makes players quit in disgust. **An entire industry pays money to patch over this exact fact**, which is a decent sign that the intuition it violates is universal.

Now the formula, which just names what you already saw

If something occurs on average **λ** times per thousand words, then in a text of **N** words the expected count is $\lambda N/1000$, and the chance of seeing **none at all** is

$$P(zero) = e^{(-\lambda N/1000)}$$

Check it against the cake. Average 1 raisin per slice, so P(zero) = e^(−1) = **0.37.** Thirty-seven slices. **The formula is the cake.**

Nothing was assumed there beyond what you already watched happen. The formula takes one input, which is how many you expect, and returns how often you get none, and **every other feature of the situation is irrelevant to it.** That is why it transfers straight from baking to a corpus without anybody having to argue that texts resemble cakes.

The clinical anchor, exact

A biopsy that was too small.

You cannot conclude absence from a sample that did not contain enough tissue for the finding to appear in. **Everyone who has signed out a report knows this**, and it is the same arithmetic.

And notice what the report says in that situation, because it is the model for what this episode is asking the project to say. It does not report absence. **It reports that the specimen was inadequate**, which is a statement about the instrument rather than about the patient, and refusing to make that statement is how a negative gets manufactured out of nothing.

Now the corpus numbers

Measured across 76 Greek and Roman texts:

BUCKET	RATE PER 1000
violence	4.76
marks	2.90
crisis	2.07
crimes	**0.95**

`crimes` **is the rare one**, planted all the way back in Season 1 episode 7. And Season 2 episode 2 already explained why it is rare rather than merely reporting that it is: **the crime stereotype is the one Girard names with nouns**, with parricide and regicide and sacrilege, and the literature describes those events with verbs. The thin bucket is thin because it was built out of the vocabulary of verdicts.

Put it through the cake:

TEXT LENGTH	EXPECTED `CRIMES` HITS	P(ZERO)
1,500 words, one Ovid myth	1.42	**0.24**
10,000 words, a tragedy	9.49	~0.000
17,180 words, the *Bacchae*	16.31	~0.000

Read the gap between the first row and the second, because the whole season is in it. At tragedy length the bucket essentially never comes up empty and the problem does not exist. At Ovid length it comes up empty a quarter of the time. **Same instrument, same lexicon, same corpus. Only the length changed**, and the instrument silently became a different instrument.

And that is the REPAIRED lexicon. The original caught `murder` 138 times out of 297 real occurrences, so its effective rate was far lower **and the zero probability far higher.** The numbers above are the optimistic version.

The mechanism, stated once

MIN takes the weakest bucket. If the weakest bucket is zero for a quarter of your texts, **MIN is zero for a quarter of your texts**, and a statistic that is zero a quarter of the time carries almost no information.

Work through what that does to a comparison. A quarter of your texts are pinned at exactly the same value regardless of anything else in them. **A HIGH text and a LOW text that both hit an empty `crimes` bucket are indistinguishable**, not because they resemble each other but because the statistic ran out of room underneath them. Averages over a column like that are averages over a floor.

> **The minimum did not fail because Girard is wrong. It failed because you cannot take the minimum of something that is usually absent.**

And notice: Season 2 episode 12 chose MIN for entirely correct theoretical reasons. The reasoning was right. **The arithmetic was never checked.** Those are two separate competences and the project had one of them, which is the ordinary condition of anybody building an instrument out of a theory they understand better than they understand the statistic.

What would count against this

1. **The Poisson assumption.** Words in a text are **not** independent; they clump by topic. **That makes the true P(zero) higher, not lower**, so the estimate is conservative and the problem is worse than stated.
2. **The lexicon being at fault rather than the metric.** Both are true at once, and episode 3 separates them.

Handing off

Episode 3 asks why nobody caught this, and the answer is a mistake so common it has a name in every measurement field.

Questions

1. **Compute P(zero) for `violence` at 1,500 words.** Why does this only bite one bucket?
2. Poisson assumes independence and words are not independent. **Does that make the estimate too high or too low?** Work it out.
3. **Where else does taking a minimum of something rare appear** in real measurement?

Collecting episode 2

The arithmetic explains the collapse. **This episode asks why nobody saw it coming**, because the answer is a mistake with a name in every measurement field and it is invisible while you are making it.

The single sentence

MIN was validated on 10,000-word tragedies and applied to 1,500-word episodes.

Texts **six times shorter.**

And nothing announced the transition. Nobody switched instruments, nobody edited the scorer, nobody wrote a line acknowledging a change of any kind. **The same code ran on a different pile of files**, which is a thing that happens in every project on every day, and the validation quietly stopped applying at the moment the files changed.

The clinical anchor, and it is the same error exactly

A reference range from the wrong population.

The assay is fine. The machine is calibrated. **The cut-off was established in adults and you are applying it to a neonate.** Nobody did anything careless and the result is meaningless.

Follow what that does to the paperwork, because it is the part that makes it dangerous rather than merely wrong. **The number comes back flagged or not flagged**, and the flag looks exactly like every other flag in the system. There is no field on the report that says "this threshold was established in a different population." The information that would rescue you is not absent because somebody hid it. **It was never part of the object being passed around.**

The character version: a boxer who trains for one weight class and fights in another. Everything he built is real. **None of it transfers.** His conditioning is genuine, his timing was drilled against genuine opponents, and the drills were calibrated to a body that will not be in front of him.

The device: the two populations, side by side

This is the whole episode in four cells, and writing them next to each other is the fix as well as the diagnosis.

	VALIDATED ON	APPLIED TO
text type	Greek tragedy	one Ovid myth
length	~10,000 words	~1,500 words
P(`crimes` empty)	**~0.000**	**0.24**
MIN behaves	cleanly, separates genres	**saturates at zero**

Nothing in the middle column is wrong. The evidence was real. It was evidence about the left-hand column.

Why it is not an obvious mistake, which is the point of the episode

The evidence for MIN was genuine.

Season 2 episode 13 showed it: on the tragedy corpus, MIN killed the *Odyssey* book 7 false positive automatically, with no special case and nobody noticing. Every Greek tragedy at MIN ≥ 0.48, every control at MIN ≤ 0.32. **Clean separation.**

That is not weak evidence. **It is exactly the kind of evidence that should justify a choice.** If somebody had asked whether MIN was validated, the honest answer was yes, and the honest answer was accompanied by a demonstration.

So identify precisely what went wrong, because "he should have been more careful" names nothing. **The evidence was about the metric's behaviour, and behaviour is not a property a metric has on its own.**

> **The mistake was assuming a metric's behaviour is a property of the METRIC.** It is a property of the metric **and the texts it runs on**, together, and those can change without anybody noticing that anything changed.

That is why the failure survives an intelligent author. A property of the metric travels with the metric, and you carry it in your head as a settled fact, correctly learned from real evidence. **A joint property of the metric and the corpus has to be re-established every time the corpus moves**, and nothing in the world reminds you, because the corpus moving is not an event.

The rule, and it generalises to every measurement you will ever build:

> **Validate a metric at the scale you intend to use it.**

And a second, separate failure hiding in the same number

Pentheus scored `crimes 0.00`, and it is worth separating the two causes because they got tangled into one result.

Cause one, arithmetic: MIN saturates at zero on short texts. Episode 2.

Cause two, coverage: Ovid describes the killing as **tearing and rending.** The bucket contained `murder`, `parricide`, `pollution`. **None of the words Ovid actually uses.**

> **That is a lexicon coverage failure, not a failure of Girard's theory**, and the two were tangled inside a single number until somebody pulled them apart.

And they cannot be untangled from the output, which is the general lesson worth carrying. **A zero is a zero.** Nothing on the face of it records whether the bucket was empty because the text lacks the thing, or because the text has the thing under a different name, or because the text was too short for a rare thing to appear. Three completely different diagnoses, one printed value, **and the value is the only object that travels downstream.**

And notice: the fix for cause two is exactly the fix that Season 2 episode 9 **refused**, because putting the tearing vocabulary into `crimes` would have been tuning the instrument to pass. **The right fix and the cheating fix were the same edit**, distinguished only by which bucket it went into.

Which puts the project in a genuinely uncomfortable position and it should be said plainly. **There is a real coverage defect here and it is still not repaired**, because the only repair available was unavailable for reasons of discipline. The honest description is not that the instrument was defended. It is that a known fault was left in place, deliberately, and the reason it was left is better than the reason for removing it.

How you would have caught it, and it costs nothing

The general fix is one line in a pre-registration, and it is worth writing down because it applies to every metric anybody ever borrows.

> **State the population the validation came from, next to the population you are about to use it on.** Not as prose. As two entries in a table, side by side, so the mismatch is visible rather than deducible.

Round 1's registration would have read: **validated on tragedies of roughly 10,000 words; applied to Ovid episodes of roughly 1,500 words.** Nobody who wrote those two lines together would have proceeded without a sentence explaining why the transfer was safe.

The reason it was not caught is that the two facts never appeared in the same document. The validation lived in Season 2's reasoning; the application lived in round 1's corpus. Both were correct. Neither was next to the other.

> That is the same shape as Season 3 episode 8's finding, where two numbers from the same round refute each other and were never printed on one page. **This project's characteristic failure is not error. It is adjacency.**

What would count against this

1. **The tragedy-scale evidence being worthless.** It is not. It is real evidence about a different population, which is precisely the diagnosis.
2. **MIN failing for a third reason nobody has named.** Possible. **Nobody has run MIN on texts of matched length to isolate it**, which would settle it in an afternoon.

Handing off

The lexicon has been revised after seeing a failure. **Episode 4 is about the one rule that makes that survivable**, and about a disclosure nobody would have noticed was missing.

Questions

1. **Two failures in one number**, one statistical and one lexical. **How would you tell them apart if you only had the number?**
2. Name a metric in your own field **used outside the population it was validated on.** There will be one.
3. **Was the tragedy-scale evidence for MIN worthless, or just narrower than it looked?**

Collecting episode 3

The lexicon was revised **after** seeing round 1 fail. Which creates a problem that has nothing to do with honesty and everything to do with logic.

The problem, stated as a question

The texts that caused you to change the instrument cannot test the change.

You looked at Pentheus. You saw `crimes 0.00`. You worked out the tearing vocabulary was missing. You added it.

> **Now score Pentheus again. Of course it improves. You built the improvement out of Pentheus.**

That is not cheating and nobody is lying. **It is circular**, and the circularity is invisible because every individual step was reasonable.

And notice how little the improvement tells you, which is the part worth being exact about. **It is not weak evidence that the lexicon got better. It is no evidence at all.** The same edit would produce the same improvement if the tearing vocabulary were a superstition, because the words were selected by reading the text they are now being scored against. The result was fixed at the moment the edit was made and the scoring run is a formality.

The clinical anchor

A diagnostic threshold derived and validated in the same cohort.

Everyone knows it is wrong. **It is published constantly**, because the cohort is expensive and you already have it and the numbers come out beautifully. The sentence in the paper is usually "the threshold was validated in an independent subset," and the subset is independent in the sense that those particular patients were held aside, in a cohort assembled at one hospital by one team.

The machine-learning version everyone knows: training set and test set. Same idea, and the reason it is drilled into every beginner is that the error is seductive rather than obscure. **A model that has seen the answers gets everything right, and it feels like competence from the inside**, which is why the discipline has to be procedural rather than attitudinal.

The character anchor

The sharpshooter who fires at the side of a barn and then paints the target around the tightest cluster of holes.

Everybody understands why the bullseye proves nothing. Now add the step that makes it this episode. **He invites you to check his accuracy by measuring the distance from each hole to the centre of the ring.**

Those measurements are real. They are careful, they are reproducible, anybody with a tape measure gets the same numbers, and **a genuinely rigorous procedure is now**

running on top of a target that was chosen to make it come out well. The rigour is not fake. It is simply downstream of the thing that decided the answer.

Rescoring Pentheus is the tape measure.

The rule

> "**No result on the old corpus can be cited as confirmation of the revision.** Those are the texts that motivated the change."

So round 2 was run on **71 texts that had never been scored with any lexicon.**

That is an expensive rule and it is worth pricing. The old corpus is right there, already fetched, already cleaned, already debugged, with two known bugs found and fixed in it. **The 71 new texts are unfamiliar, unchecked, and will contain whatever the fetcher does badly this time.** Choosing them means trading a corpus you understand for a corpus that can tell you something.

The device: the disclosure nobody would have missed

This is the part worth the episode.

Written into the pre-registration itself, before anything ran:

> "Disclosure: the v2 lexicon **was smoke-tested on the old corpus to verify the code runs. That output was seen before this file was written.** No new corpus file was scored."

Stop and consider what that is.

Nobody would have known. It is a technical check, on the old corpus, which is already contaminated anyway. It changes nothing about the result. **Omitting it would have cost nothing and no reader could ever have detected it.**

> **It is a confession of a minor contamination, volunteered, in the one document that had every incentive to leave it out.**

And there is no mechanism by which it could have been caught. This is not a case of somebody disclosing early because the truth would surface later. **Running a script to see whether it crashes leaves no trace anybody would recognise as evidence**, and the person who ran it is the only witness that it happened.

That is the actual test of whether a pre-registration culture is real. Not whether the big rules are followed, because the big rules are enforced by other people looking. **Whether the small, undetectable, harmless ones are declared anyway**, because the habit is what protects you when the stakes are high.

And be precise about why the habit matters rather than the individual disclosure. The smoke test genuinely changed nothing. **What it establishes is a threshold**: this project treats "I saw output before freezing the prediction" as reportable at all, at the level where it obviously does not matter, which is the only level where you can set a threshold honestly.

Put the ladder on screen, because the argument is about which rung you set the threshold on.

RUNG	WHAT IT IS	WHO COULD CATCH YOU	DISCLOSED HERE
5	scoring the corpus, then writing the prediction	anybody, from the git log	n/a
4	seeing a partial result before freezing	a co-author	n/a
3	reading the texts closely while drafting the lexicon	**nobody**	yes, in round 2
2	**running the code on the old corpus to see if it crashes**	**nobody, ever**	**yes, this one**
1	knowing the literature before you start	nobody, and unavoidable	stated as a limit

Read up the right-hand column. **Enforcement stops at rung 4.** Everything below it is on the honour of one person, and rung 2 is the lowest rung that could even be described as contamination.

> **A threshold set at rung 2 is only meaningful because it was set before anybody knew which rung would matter.** You cannot decide what counts as contamination once you know which way the contamination cuts.

The thing this rule does not fix, and it should be said here

The clean corpus rule protects the **instrument**. It says nothing about the **hypothesis**.

Round 2's predictions were written by somebody who had already watched round 1 fail, and who therefore knew a great deal about how these buckets behave on Greek texts. **He had seen which genres scored high. He had seen that `crimes` was thin.** None of that knowledge was erased by fetching 71 new files.

So the honest statement of what round 2 establishes is narrower than it looks. **The lexicon had not seen these texts. The person writing the predictions had seen texts like them**, and the predictions are shaped by that experience in ways nobody can quantify.

> **A held-out corpus makes the instrument honest. It cannot make the experimenter naive**, and there is no procedure that can, which is why pre-registration records what you expected rather than pretending you expected nothing.

What would count against this

1. **A smoke test being genuine contamination.** Argue it: seeing the code run on old texts tells you nothing about new ones. **The counter is that "nothing" is doing a lot of work in that sentence** and the honest move is to let the reader price it.
2. **71 texts not being enough.** A real question nobody asks. **What decides how many you need?**

Handing off

The corpus is clean and the predictions are frozen. **Episode 5 is one of the six predictions**, and it is the one nobody thinks about and everything depends on.

Questions

1. **Is a smoke test contamination?** Argue both ways.

2. **Why volunteer it at all**, when nobody would have known?
3. **How many texts do you need for a clean test?** What decides it?

Collecting episode 4

Round 2 has six predictions. Five are about Girard.

This episode is about the sixth, which has nothing to do with Girard at all, and on which every result in six seasons depends.

The prediction nobody thinks about

> **P5.** Two different translations of the **same play**, by different translators, should land within **25%** of each other on total rate.

Aeschylus' *Agamemnon*. Murray's translation, and a different one.

The device, and the clinical anchor is the same object: two labs, one sample

This is not an analogy, it is the same procedure.

You send one blood sample to two laboratories. **If they disagree, nothing either of them has ever reported about any patient means anything**, and no amount of clever analysis downstream repairs it.

Notice that this is a different kind of check from everything else in the season. Every other test asks whether a hypothesis about the texts is right. **This one asks whether the instrument is measuring the object it is pointed at**, and it has to pass before any of the others are worth running, because a failure here does not weaken the other results. It deletes them.

> The instrument is supposed to be measuring **the text.** If it is measuring **the translator**, then every comparison in this project is a comparison of two Edwardian Englishmen's word choices.

The character anchor

Two subs of the same episode.

Anybody who has watched anime with one group's subtitles and then rewatched with another's knows the experience. Same scene, same performance, same original script, **and two English lines that carry a noticeably different weight.** One renders a threat as flat and clipped, the other makes it florid. Fan arguments about which is correct run for years.

Now imagine building a tool that counts aggressive vocabulary and running it over one group's subtitles across a whole season. **You would learn a great deal about that translation team.** Whether you had learned anything about the show is a completely separate question, and the tool cannot tell you which one you got.

This project is that tool, pointed at Greek, and this episode is the only place it checks.

The result

	TOTAL
translation A	14.82
translation B	12.64
	17.2% apart

Pass.

Why this is the most important number in the project

> "Had this failed, **every cross-text comparison in the entire project would have been void**, because the instrument would have been measuring translators' word choices rather than texts."

Every single result in six seasons rests on this one control. Not the p-values. **This.**

And it earns that status by position rather than by cleverness. Every other test sits downstream of it. **A p-value tells you how surprising a number is, given that the number measures what you think it measures**, and that clause is not something statistics can supply. It has to be established separately, by a control like this one, or assumed silently by everybody.

And now the weakness, in the same episode

It has been run on exactly one paired play. n = 1.

The results file itself says it "should be repeated on more paired translations." **It has not been.**

> **The most load-bearing number in the project has a sample size of one**, and saying so is the episode.

Put those two facts next to each other and hold them there. Every result in the project depends on this control. This control is one comparison between two translations of one play. **Nothing else in six seasons rests on a single observation, and the thing that does is the thing everything else rests on.**

And there is a second thing to notice, which is worth more than the first. 17.2% is a **pass** against a 25% bound. But is 17.2% *small*?

Ask what the number actually is before answering. It is the disagreement between two careful people rendering the identical Greek. **Neither translator is wrong. Neither text has been corrupted. Nothing has gone wrong at any point**, and the instrument still reports two values a fifth apart.

> So 17.2% is not a pass mark. **It is the instrument's own wobble**, measured, and any comparison between two texts that differ by less than about 17% is inside it. **The project never states that anywhere.**

That is the useful way to hold a control result, and almost nobody does it. A control is normally read as a gate: it passed, proceed. **Read it instead as a ruler.** It tells you the smallest difference this instrument is entitled to have an opinion about, and every later result should be compared against that figure before it is compared against a threshold.

Go back to episode 8 when you get there. It puts this number next to the mythography-versus-tragedy gap, and that gap is **0.5%.**

What raising n would actually cost, since the episode keeps saying nobody did

Price it, because "should be repeated" is the cheapest sentence in science and it is worth knowing what it was buying out of.

You need a play with **two independent public-domain English translations** of the whole text, both clean enough to score. That is the binding constraint, and it is not rare: Aeschylus, Sophocles and Euripides were all translated repeatedly in the window this corpus draws from.

Then you fetch, clean and score them exactly as before. **The scorer already exists. The pipeline already exists. Nothing new has to be built.**

> Realistically: **an afternoon per pair**, and five pairs would take the most load-bearing number in the project from n = 1 to n = 6.

Say that on camera, because it changes what the omission means. **This was not blocked by difficulty, cost, or access.** It was not done because the check had already passed once, and a check that has passed stops feeling like a check.

What would count against this

1. **The two translators being unusually similar.** Both are early-twentieth century, both scholarly, both British. **A modern free translation would probably blow the bound**, and nobody has tried.
2. **17.2% being large.** It is. **The pass is a pass against a bound that was set generously**, and the bound was set by the same person who wanted the pass.

Handing off

The instrument measures texts rather than translators, **within about a fifth.**

Episode 6 is the prediction that failed, and it failed on Girard's single most important example.

Questions

1. **17.2% apart is a pass against a 25% bound. Is 17.2% actually small?**
2. **What would you do to raise n**, and what would it cost?
3. If translations differ by 17%, **what does that imply about a genre gap of

0.5%?**

Collecting episode 5

The instrument measures texts rather than translators, within about a fifth. So the comparisons are allowed to proceed.

This episode is the one where they proceed and the answer comes back wrong about the single text the whole theory is built on.

The prediction

Round 2 made a specific, sharp, unhedgeable claim about the *Bacchae*. Two parts, both written down before anything was scored. The play had to score above 0.5 on the crimes bucket, and it had to land in the top quarter of the corpus on violence.

That is a reasonable thing to demand. This is Euripides' play about a mother tearing her son apart with her bare hands, surrounded by the entire female population of a city, at the instruction of a god. If a persecution detector cannot find persecution there, it is hard to see what it is for.

What came back

The crimes bucket scored 0.52 against a threshold of 0.5. Technically a pass, by two hundredths, which is noise rather than a result and the project said so.

Violence scored 4.71 against a top-quartile cut of 6.43. Not close. A clear fail.

And overall, across seventy-two texts, the *Bacchae* came thirty-fifth. Dead centre. The most famous lynching in Greek literature sits in the middle of the table, between texts nobody would describe as being about persecution at all.

Now separate the two things, because they got tangled

Here is what makes this episode worth ten minutes rather than two.

The lexicon repair worked perfectly. Remember that Season 2 episode 9 added the tearing vocabulary, carefully, to the violence bucket rather than the crimes bucket, because that is where Girard puts it. And when you look at what fired in this play, that repair did exactly what it was designed to do. `Tore` fires seven times. `Torn` seven. `Tear` three. `Rent` twice, `rending` twice, `asunder` twice, plus the phrases `limb from limb` and `in pieces`.

That is roughly twenty-four hits that the original lexicon was completely blind to. The words are there, the instrument now sees them, and the fix was made without cheating.

And the prediction failed anyway.

So you have a repair that succeeded and a test that failed, in the same result, and if you do not pull them apart you will draw the wrong conclusion from both. The temptation is to say the repair did not work. It worked. Something else is wrong.

The mechanism, and it is arithmetic rather than literature

The *Bacchae* is seventeen thousand one hundred and eighty words long.

The killing is one messenger speech. Call it six hundred words, generously.

So the event the entire play exists to deliver occupies about three and a half percent of the text. And a rate per thousand words takes every hit in that speech and averages it across all seventeen thousand.

Think about what that does. The lynching is real, it is dense, it is exactly what the instrument is looking for, and the instrument divides it by thirty before reporting it.

Meanwhile, somewhere else in the corpus, there is a text called `quintus_smyrnaeus_2`, which is continuous battle narrative. Men fighting men, page after page, for the whole book. It scores 8.32 on violence, nearly twice the *Bacchae*.

And there is no victim in it. No crowd turning on one person. No accusation. No crisis of distinctions. It is simply violent, evenly, everywhere.

> **The instrument prefers a uniformly violent text to a text containing one lynching.** And that is not a small calibration problem. It is precisely backwards from the theory, because Girard's mechanism is not a quantity of violence spread through a culture. It is a localized event.

The device: divide it on screen

Do the arithmetic in front of them.

the *Bacchae*	**17,180 words**
the messenger speech, generously	**~600 words**
the lynching as a share of the play	**3.5%**
what a whole-text rate does to it	**divides it by ~30**

> The event the entire play exists to deliver is **three and a half percent of the text**, and the instrument averages it across the other ninety-six.

The clinical anchor, and it is exact

You are looking for a focal lesion using a serum marker.

The tumour is real. It is right there. You could put your finger on it. And the concentration in the blood comes back unremarkable, because the body is large and the lesion is small and the assay is reporting an average over the whole circulating volume.

Nobody sensible concludes the patient is well. The correct conclusion is that this assay cannot see focal disease, and you need imaging instead of chemistry.

That is exactly the diagnosis here, and it is why round 3 exists. And notice the form of the conclusion, because it is the one this whole season keeps arriving at. **The answer is not a different threshold. It is a different modality.** No recalibration of a blood test turns it into a scan, and no adjustment to a rate turns it into a statement about where something happened.

The character anchor

A film runs a hundred and twenty minutes. **The thing everybody remembers is the scene that lasts ninety seconds.**

Think of any film you would describe by one moment. Now imagine reviewing it by measuring the average intensity of every frame, and reporting the number as the film's rating. **The scene is still in there.** It has simply been averaged against two hours of people driving places and having conversations, and the number that comes out ranks it below an action film that is loud the whole way through.

Nobody would accept that as film criticism. **It is exactly what a rate per thousand words does to the* Bacchae***, and the messenger speech is the ninety seconds.

What would count against this

There are two ways this reading could be wrong and both are worth saying.

The first is that the *Bacchae* might genuinely not be a persecution text in Girard's sense, in which case the instrument is right and the theory picked a bad flagship example. Season 1 episode 15 found something that supports this: the accusation stereotype really is weak in that play, because the mob believes it is killing a lion rather than punishing a criminal.

The second is that the whole corpus might be wrong about what a top quartile means. If most of the seventy-two texts are more violent than persecution texts typically are, the threshold is mis-set rather than the play mis-scored.

Neither has been tested.

Handing off

Episode 7 is the other failed prediction from the same round, and where episode 6 was a problem with events being diluted, episode 7 is a problem with prose being dense.

Questions

1. State the difference between "this text is about persecution" and "this text contains a persecution" in one sentence.
2. If the lexicon worked and the metric failed, what should change, and how would you know you had changed the right thing?
3. Design a measurement that could find one lynching inside seventeen thousand words. You have until episode 11.

Collecting episode 6

Episode 6 was an event getting diluted across a long text. This episode is the mirror image: **a text so compressed that it looks like an event all the way through.**

Same instrument, same corpus, same round. **Opposite failure**, which is worth saying before the numbers, because a measurement that can fail in both directions at once is not going to be repaired by adjusting anything.

The prediction

Round 2 predicted an ordering of genres. Tragedy on top, then Ovid and epic and mythography clustered in the middle, then the control group at the bottom.

That is a sensible prediction. Tragedy is where persecution plots live. Control texts are hymns and genealogies with no plot at all. Everything else sits between.

What came back

The bottom half held perfectly. Tragedy at 13.88 cleared Ovid at 9.85, epic at 9.48, and the controls at 7.77. Clean, ordered, exactly as written down.

Stop on that for a second, because nobody in the project ever did. That is a four-way ordering, predicted in advance, with the gaps in the right places and no overlap. Tragedy above Ovid by four points. Ovid above epic. Epic above a control group of hymns and genealogies by nearly two. **On its own that is a clean pass**, and it is the sort of result most papers would build an abstract around.

It gets mentioned nowhere, because of what happened at the top.

And mythography came in at **13.95.**

Above tragedy. Not far above, but above, when it was supposed to be in the middle group with epic and Ovid.

One tier out of five landed wrong, and the round is remembered, by the person who ran it, as the round where the genre prediction failed. Which is correct by the letter of the pre-registration, and it is worth noticing how completely the one failure swallowed the four successes, **including in the account written by the person who wanted the successes.**

What mythography is, and why the tier is small

Two authors. **Apollodorus and Diodorus**, and eight files between them, against 36 for epic.

Season 2 episode 4 already showed where that ratio comes from, and it is not scholarship. **It is how somebody split files when they uploaded them.** So the tier that broke the prediction is the smallest one in the corpus and its size was set by an upload convention.

Hold that, because it is the second thing this episode cannot settle and the first thing episode 8 will make matter less.

The character anchor, and you already have it

The Wikipedia plot summary from Season 1 episode 17.

That episode used it to make a point about reading: whatever survives a plot summary was in the events, and whatever evaporates was in the telling. **This episode measures the summary.**

And now ask the question that decides this round. If you scored films by **incident per minute**, which would win, the film or its two-minute trailer?

The trailer, every time, and by a wide margin. **Not because more happens in it.** Because everything that is not an incident has been removed, and incident per minute is a ratio with the removed material in the denominator.

Apollodorus is the trailer. Same myths, same killings, same expulsions, with the poetry taken out.

The diagnosis the project made

The mythography texts are Apollodorus and Diodorus. They are bare plot summaries. No description, no lyric, no digression, no dialogue worth the name. A character is born, marries, kills somebody and dies in four lines, and then the next one starts.

So the reasoning went: a rate per thousand words rewards plot per word. Apollodorus is nothing but plot per word. Therefore it scores like a tragedy without being one, and the tie is **an artifact of prose style rather than a fact about persecution.**

That reasoning is good, and it was checked much later, with a **neutral** event-word list containing nothing Girardian in it at all.

GENRE	NEUTRAL EVENT-WORDS PER 1000
MYTHOGRAPHY	**28.65**
EPIC	22.68
OVID	21.73
TRAGEDY	21.45
CONTROL	20.77

It really is the densest narrative prose in the corpus, by a wide margin, and the July impression turned out to be a measurable fact rather than a story told about an inconvenient result.

The clinical anchor

A concentration reported per dry weight instead of per wet weight.

Take a sample, drive the water off, measure again. **The concentration rises, and nothing about the specimen changed.** You removed something from the denominator, which is not a discovery about the analyte.

Both numbers are correct. They answer different questions, and a table that mixes them silently produces an ordering that looks like biology and is bookkeeping.

> **Stripping the lyric out of a text is drying the sample.** The rate rises without a single additional event being described.

The prediction that was right, and why nobody counts it

One more thing before the objections, and it is the part that gets lost.

Round 2 predicted the control group would come last, and it did, at 7.77 against tragedy's 13.88. That is not a small gap and it is not a coincidence: the controls are hymns, genealogies and didactic verse, chosen in advance for having no persecution plot in them.

> **The instrument can tell a persecution text from a hymn.** That is a real capability, it was predicted, and it held.

It is also, as Season 3 episode 10 will say about a bigger number, **the uncontroversial half.** Nobody doubted that plays about people being killed use different vocabulary from hymns. The prediction was safe when it was written and it stayed safe.

Keep both facts. The instrument works, and the thing it does reliably is the thing nobody needed an instrument for.

What would count against this

1. **The genre imbalance doing the work instead.** Mythography is 8 files and epic is 36, for reasons Season 2 episode 4 showed have nothing to do with antiquity. **Nobody has reweighted and re-run**, so nobody knows whether the tie survives a corpus that is not shaped by how somebody split files.
2. **Density being the wrong explanation entirely.** It is measured, it is real, and it is still a story told after the fact about a prediction that failed. The version that would be evidence is a prediction of the density effect made **before** seeing the genre means, and it does not exist.
3. **The whole comparison being too small to discuss.** Hold that thought for exactly one episode.

Handing off

The tie has a diagnosis, and the diagnosis is measured, and the project wrote it into three seasons of documents.

Episode 8 asks whether the tie was ever big enough to need explaining, and the answer has been sitting two predictions earlier in the same round since July.

Questions

1. Mythography is the densest prose and it scored highest. **Is that a confound or a finding?** Say what would tell them apart.

2. What would you divide by, if not total words? **Name the cost of your answer**, because every denominator has one.

3. The trailer beats the film on incident per minute. **Name a measurement in your own field with that shape**, where compression is rewarded and nobody says so.

This episode is one subtraction. It needs no new data, no new instrument and no statistics beyond what you can do in your head, and it removes a confound that three seasons of documents were built to explain.

Collecting episode 7

Mythography scored 13.95. Tragedy scored 13.88.

The project called that a failed prediction, diagnosed it as a density confound, measured a neutral event-word baseline to confirm the diagnosis, and wrote the confound into every later document as a standing caveat.

All of that work is downstream of one assumption nobody stated, which is that 13.95 and 13.88 are different numbers.

Go back to episode 5

Two translations of the same play, by two competent scholars working from the same Greek, differ by **17.2 percent** on exactly this measurement.

That was reported as a pass, and it is a pass, and **it also tells you something the project never drew out.**

> **It tells you the noise floor.**

If simply changing translator moves the number by seventeen percent, **any difference smaller than that is not a finding.** It is the instrument's own wobble, measured by the instrument, and published in the same round.

Now do the subtraction

Mythography 13.95 against tragedy 13.88.

The gap is 0.07.

Divide it by the numbers it sits between and you get **one half of one percent.**

Put the two figures on screen together, because they have never appeared on the same page anywhere in the project:

the instrument's measured wobble	**17.2%**
the gap the project spent three seasons explaining	**0.5%**

> The project treated a difference of half a percent as a failed prediction, diagnosed it, wrote a confound into three seasons of documents, and computed a neutral event-word baseline to explain it. **And it sits inside a noise floor of seventeen percent, established by the project's own control, two predictions earlier in the same round.**

Say that on camera slowly. **It is the kind of thing that only shows up when somebody puts two numbers side by side that were never printed together**, and that is a repeatable move rather than a stroke of luck. Every project has a control somewhere whose result was read as a gate and never read as a ruler.

The character anchor

A photo finish decided by a stopwatch.

Two runners cross the line and the recorded times are 10.31 and 10.30. **On the face of it, a result.** One number is lower and the digits are right there.

Now ask what started the watch. A human thumb, with a reaction time of roughly two tenths of a second, **varying between attempts by more than the gap being adjudicated.** The measurement is honest, the arithmetic is correct, and the comparison is void.

Nobody in that scenario is cheating. **The thumb was never measured**, or rather it was measured, in a different session, by somebody who filed the result as a successful calibration and moved on.

The clinical anchor

Reporting a laboratory value to more decimal places than the assay supports.

The number looks precise. It is printed precisely. **And two runs on the same sample differ by more than the difference you are excited about.**

Nobody in the chain has lied. **The precision was implied by the format rather than claimed by anyone**, which is why it survives review: there is no sentence anywhere to object to.

Now run the floor against every other comparison in the round

This is the part nobody does, and it is the only reason a noise floor is worth computing. **A floor is not a fact about one confound. It is a ruler you now have to hold up against every number you have already published.**

So hold it up. The 17.2% was the difference between the two *Agamemnon* translations divided by the lower of the two, so use that basis throughout and the comparison is like for like.

COMPARISON FROM ROUND 2	GAP	VERDICT
the floor itself, two *Agamemnon* translations	**17.2%**	the ruler
tragedy against control	78.6%	far clear
tragedy against epic	46.4%	clear
tragedy against Ovid	40.9%	clear
epic against control	22.0%	clears by 4.8
named tier separation, lowest HIGH against highest LOW	**20.0%**	**clears by 2.8**
Ovid against epic	**3.9%**	**inside the floor**
mythography against tragedy	**0.5%**	**inside the floor**

Three things come out of that table and none of them was written anywhere before.

One. Two of the eight comparisons are inside the floor. Mythography against tragedy is the one this episode already knows about. **Ovid against epic is the new one**, and at 3.9% it is not a small difference, it is **not a difference**.

Two, and this is the important one. The named tier separation clears the floor by 2.8 percentage points. That is P2 and P3, the only prediction in round 2 that passed cleanly, on texts named before scoring, and the thing Season 4 episode 17 later confirms survives deleting half the `marks` bucket.

> It clears. **It clears by less than three points on a ruler measured from a single pair of translations.** The result stands and it stands much closer to the edge than anybody reading "pass" would imagine.

Three. The comfortable results are genuinely comfortable. Tragedy against control at 78.6% is not close to anything. So the floor does not dissolve the round. **It sorts it**, into three results that are safe, one that is marginal, and two that were never differences at all.

And notice what that does to the shape of P1. It predicted tragedy above **Ovid approximately equal to epic approximately equal to mythography** above control. The two approximate-equalities in the middle are the pairs now sitting inside the floor. **They came out equal, which is what was predicted, and the instrument could not have shown anything else**, so that half of P1 was never a test.

Does this make the confound disappear

No, and be careful here, because the tempting move is to say the whole thing was noise and walk away feeling clever.

The density finding is real and it is independent of this. Mythography genuinely is the densest narrative prose in the corpus at 28.65 event-words per thousand. That was measured directly, with a different word-list, and it does not depend on the 13.95 at all.

What collapses is **the evidential weight of the tie itself.** The tie is not evidence of anything. It is two numbers that are the same number, reported to a precision the instrument does not have.

> So the correct sentence is not "the density confound was imaginary." It is **"the density confound is real and this particular tie never needed it."**

And the twist that stops anyone getting comfortable

There is one more measurement, made on the same day as the event-word baseline, and it goes the wrong way for everybody.

Across all seventy-three texts, the correlation between event density and the Girard score is **negative.** Minus 0.243, with a permutation p of 0.040.

So density does not generally inflate the score. It deflates it. Epic is dense and scores low.

Which means the confound is real for mythography specifically and **unproven as a general mechanism**, and nobody has explained why mythography is an exception.

> Say plainly that the explanation does not exist. **This is an open problem in the project, not a solved one**, and pretending otherwise would be the exact behaviour this series is about.

What would count against this

1. **The 17.2% being peculiar to those two translators.** It is one paired play. **n = 1**, as episode 5 said, and with a sample size of one nobody can say whether seventeen percent is typical, generous or strict. **The entire noise-floor argument rests on that one comparison**, which is a thin foundation for an argument this destructive.
2. **A noise floor not applying to a median across a tier.** A serious objection and the best one available. The 17.2% is the wobble of a **single text** under a change of translator. Averaging over 8 mythography texts and 23 tragedies should shrink that wobble considerably, and **nobody has worked out by how much.** Do that calculation before using this argument in public.
3. **This being hindsight.** It is not. Both numbers were published in the same round, in July, by the same person. **Nothing here was unavailable at the time**, which is the uncomfortable part rather than a defence.

Handing off

Two failed predictions, both diagnosed, and one of them now disarmed by a number that was already on the page.

Episode 9 is the tool you need before the next episode, because from here this series quotes p-values constantly and has never once said what one is.

Questions

1. A gap of half a percent inside a noise floor of seventeen. **What should have been reported instead?** Write the sentence.

2. Objection 2 above is the strongest one against this episode. **Work it out.** Does averaging over a tier rescue the 0.07?

3. Every project has a control read as a gate rather than a ruler. **Find the one in your own work** and say what it would tell you if you read it the other way.

Why this episode is here

From the next episode onward this series quotes p-values constantly, and it has never said what one is. That is the same failure as a methods section that says "we tokenised the corpus."

So before the number that matters, the tool.

Start with the wrong definition, because almost everybody has it

Ask people what a p-value is and you get one of two answers. Either it is the probability that the result happened by chance, or it is the probability that the claim is false.

It is neither of those. It is not a probability about your claim at all, and once you see why, a lot of published science reads differently.

What it actually is

A p-value is the probability of getting a result at least as extreme as the one you got, **if nothing were going on.**

Everything hangs on that last clause. It is a conditional probability, and the condition is the null. You are not measuring your hypothesis. You are measuring **how surprised the null should be** to have produced what you saw.

That is a much weaker thing than people think, and it is still useful, and the gap between those two sentences is where most misreadings live.

The character anchor

A friend who says he can tell Coke from Pepsi.

You do not argue with him. You pour ten cups behind a screen and he calls nine right.

Now ask the only question available to you. **If he were guessing, how often would somebody get nine or more out of ten?** That is a question you can answer exactly, without knowing anything about him, about taste, or about soft drinks, **because it is a question about coin flips and not about your friend.**

The answer is about one time in a hundred. So you say: either he can do it, or something unusual happened. **You have not measured his palate.** You have measured how embarrassing his performance is for the theory that he is guessing, and that turns out to be the only thing an experiment ever gives you.

Notice what the test cannot do. It cannot tell you he is a supertaster rather than lucky, and it cannot tell you the effect is large: a man who is right 60% of the time forever will also produce tiny p-values, given enough cups. **The question is always about the guessing hypothesis and never about him.**

Now build one, with no formula, using the actual case

Round 2 asked whether Girard's four word-lists separate tragedy from the control group better than random word-lists would.

Step one. Measure the real thing. The buckets separate the two groups by 6.10 per thousand words. That number on its own means nothing, because you have nothing to compare it against.

Step two. Build a fake version of the world where nothing is going on. Take sixty-six words at random from the corpus, matched so they are about as common as Girard's words are, so you are not comparing rare words with common ones. That list has no theory behind it. Nobody chose it for meaning. Score it exactly the same way.

Step three. Do that two thousand times, and write down the separation each time. Now you have an entire distribution of what "nothing going on" looks like for this corpus, this metric and these two groups. It comes out centred at about minus 0.16, wandering roughly three either side.

So a meaningless list of words typically separates these groups by about zero.

Step four. Count how many of the two thousand fakes did as well as the real one.

Eight. Eight out of two thousand. Which is where p equals 0.0045 comes from.

And notice what that sentence actually says. It is not an 8-in-2000 chance that Girard is right. It is an 8-in-2000 chance that a **meaningless word list does this well**, which is a completely different claim.

The device: build the null in four steps

Write the four steps on the board and fill the right-hand column as you go.

STEP	WHAT YOU DO	ROUND 2'S NUMBER
1	measure the real thing	separation of **6.10**
2	build one fake world where nothing is going on	66 random frequency-matched words
3	do it two thousand times	fakes centre on **−0.16**, spread **2.95**
4	count how many fakes beat the real one	**8 of 2000**

The formula, now that the idea is already there

p equals the number of fakes at least as extreme, plus one, divided by the number of fakes plus one.

The plus one on both sides is worth explaining rather than skipping. Without it, if no fake ever beat you, you would report a p of zero, which claims infinite evidence from two thousand tries. The correction makes the smallest reportable value one in two thousand and one, which is honest about how much work you actually did.

The clinical anchor

You already use this every time you read a trial, whether or not anyone put it in these words.

The p-value on a treatment effect is asking: if this drug did nothing at all, how often would randomisation on its own have handed us a gap this big between the two arms? Not "is the drug real." **How often would chance alone produce this.**

The four things it is not, said one at a time

It is not the probability the hypothesis is false. It assumes the null is true and reports on the data, which is the reverse direction.

It is not the probability the result was chance. It is the probability of data this extreme **given** chance, and swapping those two around is the commonest error in the sciences.

It is not a measure of how big the effect is. A tiny effect in a huge sample gives a tiny p, which is why enormous studies report significant differences nobody cares about.

And it is not a measure of how important the finding is. Which brings us to the next episode, because p equals 0.0045 in this project means **persecution texts contain persecution vocabulary**, and that is a clean, correct, and profoundly unsurprising thing to have shown.

Where 0.05 comes from, and it is not from mathematics

Every prediction in this project is judged against **$p < 0.05$**, and the number appears so often that it starts to feel like a law. It is not one. It is a convention, and it has an author.

Ronald Fisher, in a statistics textbook in 1925, remarked that one in twenty was a convenient line for deciding whether a result was worth a second look. He was describing a rule of thumb for his own work. He was not proposing a threshold for a century of science, and he later wrote against using any fixed level at all.

> **There is nothing behind 0.05 except that it is a round fraction and somebody influential said it out loud.** One in twenty. It could as easily have been one in ten or one in a hundred, and in fields that care more, it is.

Two things follow, and both matter for reading every result in this series.

First, $p = 0.049$ and $p = 0.051$ are the same result. They differ by nothing. Treating one as a finding and the other as a failure is an artifact of a line drawn in 1925, which is why this project reports the actual p-value everywhere rather than the word "significant".

Second, a threshold only means anything if you fix it before you look. The number is arbitrary, and arbitrary is fine. **Arbitrary and chosen afterwards is not**, because at that point you are picking the line that puts your result on the right side of it.

One-sided or two-sided, and this project is one-sided everywhere

There is a fork in every p-value that almost nobody says out loud, and it is worth twenty seconds because it changes every number in this series by a factor of two.

A **two-sided** test asks *is there a difference?* and treats a surprise in either direction as a surprise. A **one-sided** test asks *is it bigger, in the direction I named in advance?* and only counts the surprises going the way you predicted.

	THE QUESTION	A HUGE EFFECT THE WRONG WAY
two-sided	is there a difference	**counts**
one-sided	is it bigger, as predicted	**returns nothing**

Every prediction in this project is directional. Tragedy above control, Gospel above myth, dissent nearer the violence. So every p-value in every round is one-sided.

> **And a one-sided p-value is exactly half its two-sided equivalent on the same data.** Anybody who recomputes these numbers the ordinary way will get figures twice as large, and the pre-registrations do not say which they used.

Say it before a commenter does. The results here survive doubling, which is the only reason it is safe to say plainly, and **that is a fact about these particular numbers rather than a defence of the practice.**

The cost is real and it is why the direction must be fixed in advance. **You buy sensitivity by agreeing not to notice being wrong backwards**, and Season 4 episode 5 contains a result that runs backwards, which is exactly the case a one-sided test is blind to.

And the thing that breaks p-values, which is coming later

If you make one prediction at a threshold of 0.05, you accept a one in twenty chance of a false pass. If you make thirty predictions, you should expect about one and a half false passes **even if nothing is true anywhere.**

This project made roughly fifty committed predictions and applied no correction for that anywhere.

It matters less than it sounds, for a reason worth working out yourself before Season 7 tells you. Here is the hint: think about what multiplicity inflates, and then look at how many of this project's predictions passed.

What would count against this episode

1. **The whole framework being the wrong one.** A serious body of opinion holds that p-values should be abandoned rather than explained, in favour of effect sizes and intervals. **This episode teaches the tool the project actually used**, and a viewer should know that teaching it is not the same as endorsing it.
2. **The permutation approach being unusual.** Most published p-values come from a formula with assumptions baked in. **The version built here, by shuffling two thousand times, assumes almost nothing** and is therefore easier to understand and harder to get wrong. That makes it a good teaching case and an unrepresentative one.

Handing off

You have the tool. **Episode 9 is the one number in the project that survives**, and the whole of its value depends on reading it with the definition above rather than the definition everybody carries around.

Questions

1. Write the definition of a p-value in your own words, with no formula, in one sentence. Then check it against the four things it is not.
2. One study reports p equals 0.049 and another reports 0.051. How different are they, really?
3. Fifty predictions, no correction. Work out why that might matter less here than in a normal study.

Collecting episode 9

You now know what a p-value is. This episode is the only one in the entire project that survives, and it is worth understanding exactly what it says and exactly what it does not.

What was tested

Tragedy against the control group. The groups were fixed in writing before any scoring happened, which matters more than anything else in this episode. Two thousand random word-sets, matched on frequency to the Girard lexicon.

The buckets separated the two groups by 6.10 per thousand words. The random sets averaged minus 0.16, with a spread of about 2.95. Eight of the two thousand beat the real lexicon.

p equals 0.0045.

Why this one is citable when two earlier ones were not

There were two null tests before this and neither can be used, and the reasons are worth stating because they are the two commonest ways a good test goes bad.

The first null test chose its list of persecution texts **after** seeing the results grid. So the groups were selected to separate, and then the separation was tested. That is circular and the project labelled it so.

The second used total rate rather than the pre-registered minimum. Switching metrics after seeing the result is exactly the move episode 1 of this season was about, and the project refused to claim it then and correctly refuses to claim it here.

This one had **both the groups and the metric committed in advance.** That is the entire difference, and it is why one number out of three survives.

And now the part that must be in the same breath

Here is what the pre-registration said this result would mean, written before anyone knew what it would be.

> "A confirmation here means only that the four stereotypes co-occur more in persecution-shaped texts than in others, **the first half of Girard's claim, and the uncontroversial half.**"

Read that twice. **Persecution texts contain persecution vocabulary.**

That is what was demonstrated. It is real, it is clean, it is properly controlled, and it is not surprising to anybody. Nobody ever doubted it. If it had failed, the instrument would have been broken.

The interesting claim, the one that makes Girard a theory rather than a vocabulary, is that the four stereotypes are a **system** that converges on one event. That is Season 1 episode 10's level three, and it is rounds 3 and 4, and they both fail.

> So the honest sentence is this: **the project's only surviving positive result confirms the half of the theory nobody disputed.**

The character anchor

A mic check.

Somebody taps the microphone and says test, one two. It has to happen, the show cannot go ahead without it, and **a failure at that moment cancels the evening.** It is genuinely load-bearing.

Nobody applauds.

That is exactly the standing of p equals 0.0045. It is a real check, correctly performed, and it had to pass. What it is not is the performance. And the danger is entirely in the reporting, because a p-value of 0.0045 written on a slide looks identical to a discovery, carries the same typography, and produces the same reaction in a room. **Nothing about the number tells you it was the mic check.**

So the useful question to carry out of this episode is not whether a result is significant. It is: **would this have been reported at all if it had passed silently?** A finding you would have shrugged at is a calibration, and it should be labelled as one by the person who ran it, because nobody downstream can tell.

The device: what the number says, and what it does not

Two columns, and the right-hand one is the episode.

P = 0.0045 DOES SAY	P = 0.0045 DOES NOT SAY
a meaningless word-list rarely does this well	Girard is right
the buckets separate tragedy from hymns	the four stereotypes are a system
the groups were fixed in advance	the effect is large
the metric was fixed in advance	anything about **where** in a text

The sting, and it belongs here rather than in Season 7

This number could not be reproduced for three weeks.

The script that computed it was run inside a working session and **never saved.** The corpus is on disk, the groups are in a frozen file, the procedure is described in prose, and the actual program that turned those into 0.0045 does not exist.

So for three weeks the project's single surviving result was a number in a document with no way to check it.

It was eventually rebuilt, across forty-eight different reasonable reconstructions, and Season 4 episode 11 is that story. The short version is that it reproduces, and that the reconstruction found something about the original that nobody expected.

The clinical anchor

A result quoted in a paper whose underlying analysis code was never deposited.

Everybody has read one. Most people have written one. The number is probably right. **And "probably right" is a different epistemic state from "checkable",** and only one of those is what the whole apparatus of pre-registration was built to produce.

What would count against this

The result would weaken considerably if the frequency matching were poor, because then the random sets are not a fair comparison. **If the fake lists are made of rarer words than Girard's, they will separate the groups less for reasons of arithmetic alone**, and the real lexicon wins a race against a handicapped opponent. Nobody has examined the matching quality.

And it would weaken if the two groups differ on something other than persecution content. Tragedy and hymns differ in a great many ways, and the null controls for word frequency but not for genre, register, or the fact that one group is drama and the other is not. **Put that concretely: a word-list of ordinary dramatic vocabulary, with nothing Girardian in it, would also separate plays from hymns**, and this test could not tell the difference between that list and the real one.

> **That second objection is not answered anywhere in the project**, and it is the strongest thing a critic could say about the one number that survived.

Handing off

The weak half of Girard's claim is established. Episode 10 is the project turning to the strong half, and it starts by admitting that everything so far has been asking the wrong question.

Questions

1. Write the sentence you would put on screen for p equals 0.0045. Then write the sentence a critic would put on screen.
2. Is "persecution texts contain persecution vocabulary" uninteresting? Argue that it is not.
3. The groups differ in genre as well as content. How would you build a null that controls for that?

Collecting episodes 6 and 10

Two things are now clear and they point the same way.

Episode 6 showed that a rate divides an event by the length of the text it sits in, so a lynching in a long play disappears. Episode 9 showed that the one thing the instrument has established is the uncontroversial half of the theory.

Both of those are the same underlying problem, and Season 2 episode 11 named it before any of this happened: **shuffle the book and the number does not change.** A rate cannot represent a moment.

So round 3 stops asking how saturated a text is and starts asking whether it contains an episode.

The reframe, stated properly

Girard does not claim that persecution texts are drenched in persecution vocabulary. He claims that at one point in the story, the crisis and the accusation and the marks and the killing all arrive **in the same place**, because one thing produced all four.

That is a claim about a passage, not about a book. And nothing built so far can see a passage.

The character anchor

Where's Wally.

There are two questions you could ask about that page and only one of them is the game. **How many people are in this picture** is answerable, precise, and completely useless. You could count them accurately, compare pages, build a table of crowd density across the whole book, and never once find him.

Is Wally on this page is a different question, and notice what answering it requires. You do not evaluate the page. **You scan it in patches**, and a patch either contains the striped shirt and the hat and the glasses together or it does not. A page with a thousand red-and-white stripes scattered across it and no Wally scores enormously on stripes and correctly returns no.

Everything the project has done until now is counting people in the picture. This episode is the first time it looks in patches, and the four buckets are the shirt, the hat, the glasses and the cane.

The statistic

Slide a window along the text. A thousand words wide, moving in steps of two hundred and fifty, so consecutive windows overlap and no passage falls between them.

Inside each window, score the four buckets. Then take the **minimum** of the four, which is the conjunction requirement from Season 2 episode 12, applied locally rather than globally. A window full of violence with no accusation in it scores zero, exactly as it should.

Then take the **maximum** of that across all the windows.

In English: **does this text contain a passage, anywhere, where all four stereotypes turn up together?** Not how much of the vocabulary the book holds. Whether such a place exists.

That is the question Girard actually poses, asked properly for the first time in the project, and it took two failed rounds to get there.

The device: the statistic in one table

Put the four settings up and reason from them.

	SETTING	WHY
window	**1,000 words**	a passage, not a book
stride	**250 words**	overlapping, so nothing falls between
inside a window	**minimum** of four	conjunction, applied locally
across windows	**maximum**	does a passage exist **anywhere**

> In English: **does this text contain a place where all four turn up together.** Not how much vocabulary the book holds.

The clinical anchor

You have stopped asking about the serum level and started asking where the lesion is.

That is exactly the diagnosis episode 6 arrived at, and it is the correct next move rather than a clever one. When an averaged measurement cannot see focal disease, you do not refine the average. **You image.**

The rule attached before anything ran, and why it matters

There is a problem with this statistic and the project stated it rather than hiding it.

The window settings, a thousand words wide with a stride of two hundred and fifty, were chosen **after** seeing round 2 fail. So they are post-hoc. Nobody derived them from theory; they were picked because they seemed reasonable.

Which creates an obvious temptation. If the numbers come back flat, you widen the window. If they come back flat again, you widen it further, and eventually something passes and you report that.

So the settings were frozen, in writing, with this attached:

> "If 1000/250 gives a null result, **that is the result.** A window size chosen because it made the *Bacchae* pass would void this round exactly as tuning the lexicon would."

That sentence is doing the same job as the refusal in Season 2 episode 9, and it is the reason round 3's failure can be believed rather than merely reported.

And notice the general form, because it is reusable and cheap. **A free parameter is not a defect.** Every measurement has some, and demanding that they all be derived from theory would stop most work from happening. What makes a free parameter dangerous is that it can be turned after you have seen the answer, in either direction, with a plausible reason available for each turn.

So the fix is not to justify the value. It is to spend it, by naming it in advance and accepting whatever it produces. The window is still arbitrary. It is arbitrary once, in public, rather than arbitrary repeatedly in private.

What this round can and cannot show

It can show whether the four vocabularies co-locate. That is a genuine advance over everything before it.

It cannot show that the co-location is Girard's mechanism rather than an ordinary property of narrative. Violent scenes contain violence words and dead kings, and a passage where a king is killed will score on three buckets for reasons that have nothing to do with mimetic desire.

> **Nothing in this round distinguishes a persecution from a murder**, and the project does not claim otherwise, and that limit should be said out loud before the result rather than after.

Handing off

Episode 11 is the most important episode in this series, and it is about the half of round 3 that is not the statistic. Because a peak means nothing on its own. **You need to know what a peak looks like when nothing is happening**, and building that is harder than it sounds and this project got it wrong four times.

Questions

1. Why take the minimum inside the window and the maximum across windows? What would maximum-of-maximum mean? Minimum-of-minimum?
2. The settings are post-hoc and were frozen. Is freezing enough? What else could you have done?
3. Predict the result before episode 13, and write down what would change your mind.

This is the keystone of the series. Four separate failures in this project are the same mistake about this one idea, and every season after this one depends on having it. It runs long deliberately.

Collecting episode 11

Episode 11 built a statistic that asks about passages rather than totals, and froze its window settings in advance so they could not be tuned.

That is half of a test. A peak of 2.4 means nothing on its own, because there is no scale, and this episode builds the other half: **the thing you put next to a number to find out whether it is large.**

The question that forces a null into existence

You measured something. You got **6.10**.

Is that big?

You cannot answer. **Big compared to what?** There is no scale. 6.10 is not large or small until something else is standing next to it, and a null is the thing you build to stand next to it.

The definition, stated once and then unpacked

> **A null is not a claim. It is a machine for manufacturing fake data that resembles your real data in every respect except the one you are testing.**

So building one is not a statistical act. It is a **modelling** act, and the only question that matters is:

1	what does my fake data **keep** from the real data?
2	what does it **destroy**?

Whatever it destroys is what you are testing. Nothing else. If it destroys two things, you are testing two things and you will not be able to say which. If it destroys nothing that matters, you are testing nothing.

> And that last case is the series' refrain arriving in algebra. **A null that destroys nothing that matters is a check that cannot fail, and a check that cannot fail prints reassurance.** It will return a comfortable number every time, on any data, for any hypothesis.

The clinical anchor, and it is not an analogy, it is the same thing

A control group.

Same age distribution, same sex ratio, same comorbidities, same hospital, same season, same everything **except the drug.**

Now ask the two questions of it. It keeps: age, sex, comorbidity, setting. It destroys: exposure to the drug. **So it tests the drug.**

And then the two ways it goes wrong, which are exactly the two ways every null in this project went wrong.

Match on too much. If you also match on the drug, the groups are identical and you learn nothing. Less obviously: if you match on something **downstream of the drug**, you have controlled away the effect you were looking for. That is round 4's failure and round 7's, described in advance.

Match on too little. If the control group is younger, you are testing age, and the write-up will say you were testing the drug. That is round 3's second null.

> **A badly built null does not weaken your study. It silently answers a different question than the one you asked, and the write-up will not know.**

The character anchor

A boxing weight class, and you have met one already. Episode 3 used it for a fighter who trains at one weight and competes at another, which was a point about validation. **This is the same object doing a different job**, and it is worth noticing that a weight class can do both, because a control and a reference population are the same idea seen from two sides.

The entire point of a weight class is to hold constant everything that is not skill, so that what remains is skill. Nobody thinks a heavyweight beating a flyweight demonstrates technique.

And the failure mode is the interesting bit: **if you also matched on reach, stance, age, camp, and record, the fight tells you nothing**, because you have matched away the thing you were trying to see.

Now build one, properly, for this project

The question in round 3 was: **do the four stereotypes converge on one passage?**

So the fake data has to be a text that has everything the real text has, except that nothing is anywhere in particular.

The shuffle null. Take the text. Put every word in a hat. Draw them out in random order. That is your fake text.

IT KEEPS	IT DESTROYS
every single word, exactly	the order they were in
every bucket's total count, exactly	which words are near which
the whole-text rate, **to the last decimal**	any passage, any scene, any moment
the text's length	

> **So it isolates position and nothing else.** And notice what follows: a text can be absolutely drenched in all four stereotypes and still fail this null, because the shuffled version is drenched too. **The null is immune to the density confound that wrecked round 2.** That is not luck. It is what a well-chosen null buys you.

The one implementation detail that decides the whole thing

The lexicon contains multi-word phrases: `limb from limb`, `cast out`, `with one voice`.

If you shuffle the raw words, **every phrase dissolves**. `limb from limb` becomes three words scattered across the book, and the fake text scores far lower than the real one on phrases **for a reason that has nothing to do with position.** Your null would be testing "do phrases exist" and you would report it as localization.

The fix: mark each phrase hit at its **first word**, and shuffle the marks rather than the words. A phrase moves as a single unit.

> One line of code, and without it the round measures the wrong thing and looks like a triumph.

The arithmetic that comes out

Run the shuffle five hundred times per text. You now have five hundred fake answers. Compare:

$$z = (\ \textit{observed} - \textit{mean of the fakes}\ )\ /\ \textit{standard deviation of the fakes}$$

z is "how many typical wobbles away from nothing am I." z = 0 means you look exactly like nothing happening. z = 3 means you are three wobbles out.

$$p = (\ \textit{fakes at least as extreme} + 1\ )\ /\ (\ \textit{fakes} + 1\ )$$

Same plus-one correction as episode 9, same reason.

The four ways this project got it wrong, listed now and paid off later

Keep this table on screen for the rest of the series. **Every entry is the same error**, and the error is answering question 2 wrongly.

#	ROUND	WHAT THE NULL DESTROYED	WHAT IT SHOULD HAVE DESTROYED
1	3, Null B	word identity **and** clumpiness together	word identity only
2	4, D4	only the unigram half, leaving the phrases untouched	the whole instrument
3	5	a rare phrase, replaced by a word 316 times commoner	a phrase of comparable rarity
4	7	position of the dissent **and** the clustering of the violence	position of the dissent only

Four separate occasions. Four different rounds. Four different people-hours of care. The same question answered wrongly each time.

> And that is why this episode is the keystone. Not because nulls are hard mathematically, they are not. **Because deciding what counts as nothing happening is a judgement about your subject, and it is made once, early, in a line of code nobody revisits.**

The exercise, and do it before the next episode

For each of these, say what it keeps and what it destroys. Write both columns.

NULL	KEEPS	DESTROYS
shuffle the words in a text		
shuffle the group labels between texts		
draw random words matched on frequency		
keep the violence, shuffle only the dissent		
swap in a different translation of the same text		

The fourth row is the one round 8 finally used, and it is the corrected version of the fourth failure above. The fifth row is a control this project ran twice and it is the reason any of its comparisons mean anything.

What would count against this episode

1. **Some nulls being safe by construction.** If a null design existed that could not mismatch its statistic, the whole "judgement about your subject" framing is overblown. **Question 3 below asks you to find one**, and this project's four failures are evidence rather than proof.
2. **The four failures having four unrelated causes.** Read them again and decide whether "the null destroyed the wrong thing" is a real common cause or a frame imposed afterwards on four separate mistakes. **It is imposed afterwards.** The question is whether it is also true.
3. **The shuffle null being wrong here too.** It destroys all narrative structure, not only the position of the buckets, so a text must beat its own coherence and not merely chance. **Episode 13 raises this and nobody has tested it.**

Handing off

You have the two questions, and a null for round 3 built to answer them.

Episode 13 is what came back, and the answer is that word order turns out to be irrelevant to the statistic, which is the cleanest way a well-built test can tell you that you measured nothing.

Questions

1. Design a null for "this author uses short sentences at emotional moments." State what it keeps and destroys before writing any procedure.

2. A null that destroys two things at once. Why can no amount of downstream statistics repair it?

3. Is there such a thing as a null that is safe by construction, or is every one a judgement about the subject?

4. The shuffle null keeps the whole-text rate exactly. Explain in one sentence why that makes it immune to round 2's density problem.

Collecting episode 12

You have a statistic that asks about passages, and a null that destroys position and nothing else. Everything is finally set up to ask Girard's actual question.

What came back

Five predictions were committed. One passed and it passed for the wrong reason.

Nine texts had been named in advance as the ones where the mechanism should be visible. The prediction was that at least seven of the nine would localise, which means their real arrangement beats their own shuffled version.

None of them did. Zero of nine.

The *Bacchae* specifically was predicted at p below 0.01. It came in at **0.66**, which is not a near miss, it is the middle of the distribution.

Tragedy was predicted to exceed epic by at least a full standard unit. The gap came in at 0.30.

And the second null, which we will come to in the next episode, produced zero of nine as well.

The number that tells you what actually happened

Across all seventy-one texts, the z-score had a mean of plus 0.07, a spread of 1.33, and a **median of exactly zero.**

Four texts out of seventy-one cleared the significance threshold. Chance predicts 3.6.

And not one of the four was a named persecution text. They were two books of Ovid, a Nonnus and a Valerius Flaccus, which is what you would expect if you drew four texts out of a hat.

> So the conclusion is not that the effect is weak. It is that **there is no effect being measured at all.** The statistic is reporting how much bucket vocabulary a text contains, and word order is irrelevant to it.

Read the median again, because it is the single most informative number in the round. **Zero.** Not small, not near zero. The typical text in this corpus does precisely as well arranged as Euripides left it as it does with every word poured out of a bag. Half of them do worse.

Which means the apparent convergence in round 2's exploratory version was exactly what you get when a text with these vocabularies at these densities is cut into windows. Somewhere, by chance, a window holds a few of each.

And that is what a properly built null is for. **The convergence is real and it is also what nothing looks like**, and no amount of staring at the real text would have told you that, because the real text does contain the passage you thought you found.

The device: the whole round on one line

	PREDICTED	OBSERVED
named persecution texts localising	**7 of 9**	**0 of 9**
the *Bacchae*	$p < 0.01$	**$p = 0.66$**
tragedy over epic	≥ 1 standard unit	0.30
texts clearing significance	—	4, against **3.6 by chance**
median z across 71 texts	—	**exactly 0.00**

Why the one passing prediction is not a success

The prediction about the control texts was that at most one of six would localise. Zero of six did. Technically a pass.

And the project refused to count it, in its own results file:

> "A prediction that these texts show no effect is worthless in a corpus where **no text shows the effect.**"

That is worth pausing on, because it is a small act of discipline that most people would skip. It is a free pass sitting on the table. Taking it would move the scoreboard from one-of-five to two-of-five and cost nothing, and a reader skimming the results would never notice.

> **A pass that would have happened whatever the truth was is not evidence.** It is the same structure as the confirmation table in Season 2 episode 1, at the level of a single prediction.

The Bacchae, twice now

Round 2 ranked it thirty-fifth of seventy-two and the diagnosis was that the metric was wrong, because a rate dilutes one lynching across seventeen thousand words. That diagnosis was correct and the fix was built.

Asked at episode scale, with a proper null, it comes out at **p equals 0.66.**

And here is the detail that should stop anyone getting comfortable in either direction.

Its peak window sits at **fifty-four percent of the text.**

That is exactly where the `sparagmos` is. Season 1 episode 7 gave you the word: the tearing-apart, the ritual dismemberment of a living victim by a crowd. In this play it is the scene where Agave and the women of Thebes kill Pentheus with their hands.

The locator is pointing at the right passage. Not approximately. It found the one scene the entire theory is built on, without being told where to look.

That passage simply does not contain a convergence of the four stereotypes beyond what the play's own vocabulary produces by chance.

> The project's own words: "Girard's central example has now failed two pre-registered tests on two different operationalizations. The objection this project exists to answer is that the exemplars are chosen by interpretation rather than by any property the text carries, and **after two rounds that objection is stronger, not weaker.**"

The clinical anchor

An imaging study that localises the lesion correctly and cannot characterise it.

You are looking in the right place. The scan is pointing at the right organ. And the finding does not distinguish the thing you are looking for from the ordinary tissue around it, which is a different failure from looking in the wrong place and needs a different fix.

The character anchor

A sniffer dog that alerts on the right suitcase, and alerts on every other suitcase too.

The alert on the right bag is not fake. The dog really did sit down in front of the one with the contraband in it, and if that were the only bag you had shown him, you would be impressed and you would be entitled to be.

The problem only appears when you count the rest of the carousel. He alerts on all of them, so the alert on the right one carries no information, and the handler watching a single bag has no way to see that. **You cannot evaluate a detector by watching it detect.** You have to know its rate of alerting when there is nothing there, which is the entire job of the null, and it is why the peak sitting at 54% of the *Bacchae* is not evidence of anything at all.

What would count against this reading

The statistic might be too blunt to see an effect that is really there, and episode 15 is entirely about that possibility with a number attached.

And the null might be too strong. Shuffling destroys **all** structure, including ordinary narrative structure that has nothing to do with Girard, so a text has to beat not just randomness but its own coherence. Nobody has examined whether that bar is fair.

Handing off

Episode 13 is the second null from this round, and it did not fail. **It could never have succeeded**, and the reason is structural rather than empirical.

Questions

1. The locator points at the right passage and the statistic says nothing is there. Can both be true?
2. Show from the definition why word order is irrelevant to this statistic.
3. Two failures for the Bacchae. At what number do you stop defending it?

Collecting episode 12

Episode 12 gave you the two questions you ask of any null. What does it hold constant, and what does it destroy. This episode is the first of four times this project answered the second question wrongly.

What the second null was for

Round 3 had two nulls, and they were asking different things.

The first, the shuffle, destroyed **position** and held the words constant. That one was well built and episode 13 reported what it found.

The second was supposed to test **word identity**. The question is fair: maybe any four word-lists of roughly the right commonness would converge on passages the same way. Maybe there is nothing special about Girard's particular words.

So: build four random lexicons, matched to the Girard buckets on how many forms they have and how common those forms are, and score them the same way.

What came back

The random lexicons scored **higher** than Girard's.

For the *Bacchae*: the real instrument scored 2.0. The random lexicons averaged **4.41.**

Not marginally higher. More than twice as high.

Why, and it has nothing to do with Girard

Here is the mechanism, and it is worth going slowly because it is entirely structural and it recurs.

The statistic is a **minimum** over four buckets inside a window. So a window scores only as high as its weakest bucket.

Now think about what kind of word-set has a high minimum.

A set of words that is **spread evenly through a text** will have some presence in nearly every window. Its minimum across the four is decent everywhere, because nothing ever drops to zero.

A set of words that is **clumped** will be dense in a few places and absent everywhere else. So in most windows at least one of the four is at zero, and the minimum collapses.

Now: what does a random frequency-matched lexicon look like? You have drawn words from all over the language. They are about a boat, a hillside, a colour, a grandmother. **They have nothing to do with each other**, which means they appear at fairly uniform rates throughout any text.

And what does Girard's lexicon look like? Four thematically coherent sets, which by definition cluster where the theme is.

> **So the random sets win by construction.** A minimum rewards evenness, random frequency-matched words are even, and any thematically coherent lexicon is clumpy and loses.

The device: keeps and destroys, for the null that failed

Run episode 12's two questions on Null B and the failure is visible before any data arrives.

	NULL B
keeps	the number of word-forms, their corpus frequency
destroys	thematic coherence, and therefore clumpiness
so it tests	whether your lexicon is spread out
what you wanted to test	whether **these particular words** matter

And that is what makes it worthless rather than merely wrong

This is the point to say hard on camera.

Null B would produce this result for **any** four topical word-sets. A lexicon of sailing terms. A lexicon of colours. A lexicon of legal vocabulary. Every single one would lose to random words on a minimum statistic, for reasons that have nothing to do with whether the topic is present in the text.

> **A test that returns the same answer regardless of the truth is not measuring anything.** It is Season 2 episode 1's confirmation table with the sign flipped: instead of never returning a negative, it can never return a positive.

Which is the refrain, and this is the clearest instance in the project. A check that cannot fail prints reassurance, and the sign does not matter: an instrument that always says yes and an instrument that always says no are the same instrument, and neither of them is looking at anything.

The project's own results file records it in four words: **"my error, in the pre-registration."**

The clinical anchor

A control group matched on a variable that lies on the causal pathway.

Nobody sabotaged anything. The matching was careful and well intentioned. And by matching on the thing the exposure works through, you have guaranteed no difference will appear, and the study will report a null that means nothing at all.

The failure is in the design, and no amount of downstream statistics repairs it, because the data that would answer the question was never generated.

The character anchor

A pub quiz scored on your worst round.

Four rounds: history, sport, music, film. The rule is that your team's score is whatever you got in the round you did worst in.

Now enter two teams. **Four historians**, who are magnificent on history and score zero on the other three. **And four strangers pulled off the street**, none of whom knows much about anything, who each pick up two or three in every round.

The strangers win, and they win by a distance, and they will win **every single time you run it.**

Nothing about that result is a discovery about history. **It is a discovery about the scoring rule**, which rewards knowing a little about everything and punishes knowing a lot about one thing. The answer was fixed the moment somebody decided to score the worst round.

Girard's lexicon is the historians. Four thematically coherent sets, each brilliant where its theme is and absent everywhere else. **The random frequency-matched lexicon is the strangers**, drawn from all over the language, never absent anywhere, never zero in any window.

The pattern, and start counting on screen

This is the **first of four**, and keeping a tally across the series is worth doing because the pattern is the most transferable thing here.

Round 3's Null B favours evenly spread sets over clumpy ones by construction.

The other three arrive in Season 4, and each one is the same mistake in a new costume: a null that destroys something the statistic depends on, or preserves something it should have destroyed.

> **Every one is a mismatch between what the null holds constant and what the statistic is sensitive to.** Four separate rounds, four separate designs, four separate people-hours of care, and the same question answered wrongly each time.

Could it have been caught in advance

Yes, and that is the uncomfortable part. You do not need any data to see this.

You need only to ask: what does a minimum reward, and are my random draws going to have more of that than my real lexicon does. Both halves are answerable at the whiteboard, before a single line of code runs.

> **Nothing about this failure required the experiment**, and the experiment was run anyway, and it produced a number that was then reported and had to be retracted.

What would count against this episode

1. **Null B being salvageable.** If the minimum were replaced with something that does not reward evenness, the same random-lexicon comparison would be a fair test of word identity. **That is a real repair and nobody made it**, so the question round 3 asked about word identity is still unanswered rather than answered negatively.
2. **The reasoning here being hindsight.** It is not, and that is the uncomfortable part: **the argument needs no data at all**, only the question of what a minimum rewards, and it was available at the whiteboard.
3. **Thematic lexicons not actually being clumpier than random ones.** It is asserted here and it is measurable. **Nobody measured it**, so the mechanism is argued rather

than demonstrated.

Handing off

Round 3 produced one flat result from a good null and one worthless result from a bad one.

Episode 15 asks the question that decides what the flat result means, which is whether the round could have detected an effect that was really there.

Questions

1. Before Season 4: predict what the other three null failures look like.
2. Could Null B have been caught before running? By what specific check?
3. Is there such a thing as a null design that is safe by construction, or is every one a judgement about the subject?

Collecting episodes 13 and 14

Round 3 found nothing. One null was well built and returned a flat result, and the other could never have returned anything.

So the obvious question: **does "we found nothing" mean "there is nothing", or does it mean "we could not have seen it"?**

Those are completely different conclusions and they look identical in the output.

The number

The statistic is a minimum over four counts inside a thousand-word window.

Across the corpus, its observed values run from zero to five. **The median is two.**

For sixty-four of seventy-one texts, the weakest bucket contributes **three hits or fewer** in the best window in the entire text.

> **This is a statistic built on counting to two.**

Say the sentence and then sit in it. The entire round, five predictions, seventy- one texts, two nulls and two thousand permutations, rests on whether a number that is usually two is bigger than another number that is usually two.

What that means for the result

It means round 3 cannot distinguish between two situations.

Situation one: there is no localisation. The four stereotypes genuinely do not converge on passages, and the flat result is the truth.

Situation two: there is localisation, and an instrument whose values run from zero to five is far too coarse to detect it.

Both produce the same output. Both produce a median z of zero. And nothing in the round separates them.

The device: the resolution of the instrument

the statistic	minimum of four counts in a 1,000-word window
its observed range	**0 to 5**
its median	**2**
texts where the weakest bucket contributes ≤3	**64 of 71**

> **A statistic built on counting to two.** Five predictions, seventy-one texts, two nulls and two thousand permutations rest on whether a number that is usually two is bigger than another number that is usually two.

The confession, which is the reason this episode exists

From the project's own results file:

> "The honest position is that the test as specified had little power, and that **this was foreseeable before running it and was not foreseen.**"

Read the second half again. Not "we could not have known." **We could have known, by arithmetic, in advance, and we did not check.**

That is a harder sentence to write than "the test failed", and it is the correct one.

The tool: power, in one idea

Power is the probability that your test finds an effect that is really there.

A test with low power that comes back negative has told you almost nothing. And the negative looks **exactly** like a real negative, on the page, in the abstract, in the summary table. There is no visual difference between "we looked carefully and it is not there" and "we looked with a blunt instrument."

And notice that power is a property you can compute **before** collecting anything. It depends on how big an effect you are hunting, how much noise there is, and how much resolution your instrument has. **All three are knowable in advance**, which is what makes the confession above the right confession: this was not bad luck, it was an unrun calculation.

The character anchor

Weighing a letter on a bathroom scale.

You want to know whether it is over fifty grams. You put it on, and the display says **zero**.

The letter is not weightless. Nothing has malfunctioned, the scale is accurate, and the reading is what that instrument reports for every letter that will ever be placed on it. **The zero is not a measurement of the letter. It is a measurement of the scale**, and there is nothing on the display that says so.

Now imagine weighing four hundred letters this way and reporting that letters have no meaningful weight. **Every individual reading is honest**, the sample is large, the procedure is consistent, and the conclusion is manufactured entirely by choosing an instrument whose smallest step is bigger than the thing being measured.

A statistic whose values run from zero to five, with a median of two, is a bathroom scale.

The clinical anchor

An underpowered trial reporting no difference between arms.

It is the commonest error in the medical literature. Absence of evidence presented as evidence of absence, and **the abstract never tells you which one you are reading.** You have to go and compute it yourself from the sample size, and almost nobody does.

And there is a second-order effect worth knowing, because it inverts the usual intuition that a weak study is merely a harmless one. **In an underpowered study, the results that do reach significance are the ones that got lucky**, so they are systematically larger than the truth. A field full of small studies does not produce a blurry picture of reality. It produces a confident picture of an effect bigger than the real one.

The rule that follows, and it costs something

The temptation after a null result on a coarse statistic is obvious. Widen the window. More words per window means more hits per bucket, the minimum stops saturating, and eventually something reaches significance.

> **That is forbidden**, and the project forbade it in advance: no re-running with a wider window until something passes. A different operationalization is a new pre-registration with its own commit, judged on its own predictions.

Which means round 3's failure has to be lived with rather than fixed, and the next attempt starts from scratch with its own written predictions.

And the fix that eventually came

Later rounds carry a **power falsifier**: a number fixed in advance which, if crossed, declares the round **under-powered rather than negative.**

That is a genuinely good invention and it exists because of this episode. Season 4 is the first time it fires, and when it does, the round is reported as "we could not have seen it" instead of "it is not there."

> **Round 3 could not make that distinction. Every round after it can**, and the reason is that somebody wrote down a number in advance instead of arguing about it afterwards.

What would count against this episode

1. **The power problem being unfixable rather than unnoticed.** If no operationalization of this claim could have had power on texts this size, then "foreseeable and not foreseen" is too harsh and the honest verdict is that the corpus was never big enough. **Nobody has worked out which it is.**
2. **The median of two being the wrong diagnostic.** Power depends on the effect size you are hunting as well as the instrument's resolution, and **no effect size was ever specified**, so strictly the power calculation could not have been done as described. That is a worse finding than the one in the episode.
3. **The power falsifier being gameable.** A number fixed in advance can still be fixed generously. **Season 4 is where it fires**, and whether it was set at an honest level is checkable there rather than here.

Handing off

Round 3 is finished and its verdict is that nobody knows.

Episode 16 is the round that finally asks about dissent, which is the thing Girard actually claims, and it opens with a confession written before any data existed.

Questions

1. How would you have computed the power of this statistic before running it? You have everything you need.
2. Write the sentence distinguishing "no effect" from "no power" for a general audience, in one line.

3. Why is re-running with a wider window forbidden, when a wider window is obviously better?

Collecting three rounds

Three rounds have now measured the four persecution stereotypes and none of them reached anything Girard would call interesting.

And there is a reason for that which was known from the very beginning and written down in July, before any of it: **the four buckets can never separate a Gospel from a myth.**

Girard predicts the Passion scores high on all four. The Gospels are built out of the same persecution material as myth. So a high score there confirms the first half of his claim and discriminates nothing.

> **That is the design, not a defect.** Season 1 episode 13 said it. The project said it in July. **And it means every round so far has been testing the uncontroversial half on purpose**, while the interesting half sat untouched.

Round 4 is the interesting half.

The claim, finally

I See Satan Fall Like Lightning, page 3, which Season 1 episode 12 read twice:

> "in mythology, **no one ever questions this guilt**. In the Gospels, the revealing account of scapegoating emanates not from the unanimous crowd but from a dissenting few. [...] **In mythology no dissenting voice is ever heard.**"

A universal negative. **One genuine counterexample kills it.**

That is the most falsifiable shape a claim can take, and it is the shape this project has wanted since round 1. Everything before this has been comparative: this tier scores higher than that tier, on average, by some margin, and every comparative claim comes with an argument about whether the margin is big enough. **"No dissenting voice is ever heard" comes with no such argument.** Produce one myth in which somebody objects during the killing and the sentence is finished.

Which means the danger has moved. **The risk is no longer that the claim is unfalsifiable. It is that the instrument is not competent to falsify it**, and those two look identical from the outside when nothing turns up.

The instrument, and the split that decides everything

A fifth bucket, built out of two halves, and the halves are tagged separately.

The **core** half traces to page 3, the questioning of guilt. Innocent, guiltless, no fault, done nothing, unjust, false witness, protest, object, denounce, defend, plead.

The **unan** half traces to page 122, the broken unanimity. Division, dissent, some said, others said, not all, withheld consent, spare him, minority.

And every single result is reported **twice**, once with the second half and once without, which is the tagging discipline from Season 2 episode 10 applied to a new problem.

The device: the fifth bucket, split before it was scored

HALF	TRACES TO	VOCABULARY	FLAGGED IN ADVANCE?
core	*I See Satan* **p. 3**	innocent, no fault, done nothing, protest, defend, plead	no
unan	*I See Satan* **p. 122**	division, some said, others said, not all, spare him	**yes, as possibly motivated**

Every result is reported twice, once with `unan` and once without. That is the tagging discipline from Season 2 episode 10, applied before anybody knew which half would carry anything.

The exclusions, each of which makes the hypothesis harder to confirm

This is worth listing, because the direction matters.

`Have mercy` was excluded, despite being frequent in the Gospels, because it is almost always a petition for healing rather than dissent from a verdict.

`Rebuke` was excluded, because in this corpus its dominant sense is Jesus rebuking demons and wind and disciples.

Bare `witness` was excluded, because bearing witness is Johannine testimony, not objection. Only `false witness` survived.

`Condemn` was excluded, because that is what the **crowd** does, not the dissenter.

> **Every one of those removals takes hits away from the Gospel tier**, which is the tier the author of the lexicon wanted to score highest. Same move as the 960 deletions in Season 2 episode 9, and the same evidence value.

The disclosure, and this is the episode

Written into the pre-registration, before anything was scored.

> "**I knew before writing this that John repeats a division formula** and that the Passion contains 'I find no fault in this man.' Those are the two most obvious surface forms for the construct, so they **are at risk of being motivated rather than derived.**"

And then, about a prediction he was registering anyway:

> "**I expect D6 to fail**, and it is registered anyway. Stating the expectation in advance is the point."

Stop and consider what that second quotation is.

It is a researcher writing down, before the data exists, that one of his own predictions is going to fail. Not hedging. Not omitting it. **Registering a prediction he expects to lose, and saying so, so that when it loses nobody can claim it was a surprise retrofitted into a story.**

The clinical anchor

Declaring your conflict of interest **before** the study rather than in small print afterwards.

The declaration does not remove the conflict. Nothing removes the conflict. What it does is **let the reader price it**, and pricing is all a reader can ever do.

The character anchor

A talent-show judge who says in the first round that one of the contestants used to be his student.

He is still biased. Saying it changes nothing inside his head, and every generous score he gives that contestant is going to be suspect in exactly the way it was always going to be suspect.

What it changes is what a disputed score means afterwards. Nobody can produce the connection as a revelation, because it was on the record before anybody sang, and the argument moves from whether he is compromised to whether this particular score was wrong. That is a much smaller and much more answerable argument.

Now notice the version that does not work, which is the one almost everybody does: he judges the whole competition, and afterwards, once his student has won, he mentions that he taught her. Same fact, same words, **and it is worthless, because by then the disclosure is competing with a result.**

The pre-registration here names the two most obvious surface forms in the lexicon as possibly motivated, **before knowing whether they would carry anything.** By episode 18 they carry everything, and the warning is already in the file.

> And the test of whether the disclosure was real is what happens next, which is episodes 17 and 18. **The thing he warned about is exactly what the data did.**

What would count against this round before it runs

If the `unan` half carries the result and the `core` half does not, then the effect belongs to the vocabulary flagged as contaminated, and the round cannot be claimed however good the p-values look.

That is not hindsight. **It is what the sensitivity analysis was written to detect, and it was written before anybody knew.**

Handing off

The instrument is built, the contamination is disclosed in advance, and two sensitivity analyses are loaded and pointed at the result.

Episode 17 is the round running, and for the length of that episode it looks like the first thing in this project that worked.

Questions

1. Disclosing that your lexicon may be motivated does not make it unmotivated. What does the disclosure actually buy?
2. Registering a prediction you expect to fail. Cynical move or real discipline?
3. Which of the two halves do you expect to carry the effect? Commit now.

Film this one straight, with no irony in the voice. The audience has watched three rounds fail and they are owed a round that does not, and for the length of this episode they are going to get one. **What happens to it happens in episode 18**, and the episode is worth nothing if you telegraph it.

Where we are

Round 4 has just run. It is the round aimed at the thing Girard actually claims, the dissenting voice, and it is **the first round in the whole project that looks like it worked.**

The headline, and let it land

TIER	N	MEDIAN RATE PER 1000
GOSPEL	4	**1.387**
TRAGEDY	23	0.667
MYTH	71	0.547

Look at that ordering. Myth at the bottom, tragedy in the middle, the Gospels at the top, and the Gospels at roughly **two and a half times** the myth rate.

That is the ordinal series from Season 1 episode 13. The one Girard predicts. The one this project has been chasing since round 1 and has never once seen.

And the tests agree:

	RESULT
D1, the medians in the predicted order	**PASS**
D2, Gospel above myth	**PASS**, p = 0.00034
D3, tragedy above myth	**PASS**, p = 0.032
D4, the null test	FAIL

Three of four. And read D2's p-value again. **Three ten-thousandths.**

Why this one deserved to be believed

Go through what is different about round 4, because the case for it is genuinely strong and the audience should feel the strength before anything happens to it.

It asks the right question. Rounds 1 to 3 measured the four stereotypes, which Season 1 episode 13 established can never separate a Gospel from a myth, because both are built out of the same persecution material. **This round measures dissent**, which is the only place Girard says the difference lives.

The sample is bigger, at 98 texts against 71.

The tiers were frozen in a file before scoring, and the file is in the repository.

And the exclusions all cut against the hypothesis. `Have mercy` was removed despite being frequent in the Gospels, because it is a petition for healing rather than dissent from a verdict. `Rebuke` was removed, because its dominant sense here is Jesus rebuking demons and wind. Bare `witness` was removed as Johannine testimony. **Every one of those takes hits away from the tier the author wanted to win.**

> If you stopped here you would have a result. The first positive finding on the interesting half of Girard's claim, on 98 texts, predictions committed in advance, exclusions running against the author's own interest. **You could publish that. People do publish that.**

The character anchor

A boxer who tells the judges in advance which round he has to win in.

Anyone can say afterwards that the fight was close. **Naming the condition first, publicly, in a form that can go against you, is the only version that costs anything.**

Hold that image, because the rest of this episode is somebody doing exactly that, and episode 18 is the bell.

The two things that were written down before anybody looked

When round 4 was pre-registered, **two extra analyses were declared.** Not predictions. **Sensitivity analyses.**

A sensitivity analysis is a rerun with **one choice changed**, to find out whether the result depends on that choice or on the world. Every study contains dozens of choices that could have gone another way, and most of them are never revisited, because revisiting them is unpaid work that can only cost you something.

The clinical anchor

The pre-specified sensitivity analyses in a trial protocol.

You know the standard ones: different imputation for missing data, per-protocol alongside intention-to-treat. **Analyses chosen, in practice, because they will probably agree.** They are declared in advance, they are entirely legitimate, and almost none of them is capable of overturning anything.

The question to ask of any sensitivity analysis is not whether it was pre-specified. It is whether it could have gone the other way, and for most of them the honest answer is no.

And why these two were written

They exist because of something disclosed in the same document. The lexicon had two halves, tagged separately per Season 2 episode 10, and **one of them was flagged in advance as at risk of being motivated.**

The `core` half traced to Girard's page 3, the questioning of guilt. The `unan` half traced to page 122, the broken unanimity. And the pre-registration says, before any scoring:

> "I knew before writing this that John repeats a division formula and that the Passion contains 'I find no fault in this man.' Those are the two most obvious surface forms for the construct, so they are **at risk of being motivated rather than derived.**"

Read what that is. A researcher writing down, before the data exists, the exact mechanism by which his own instrument might be cheating, naming the two specific phrases, in the document that freezes his predictions.

S1 was the rerun with that half deleted. S2 was the rerun with epic excluded from the myth tier.

What would count against this round, stated before the answer

This is the honest version of the cliffhanger, and it is better than suspense because the audience can reason along.

1. **If the `unan` half carries the result and `core` does not**, the effect belongs to the vocabulary flagged as contaminated, and the round cannot be claimed however good the p-value looks.
2. **If the myth tier's median is being dragged down by epic**, then the tragedy-above-myth half of the series is a fact about epic rather than about Girard.
3. **D4 already failed**, and D4 was the null test. A round that passes three comparisons and fails its null has not yet been tested against chance in the way the other three rounds were.

None of that is hindsight. Items 1 and 2 are the two analyses, written in the pre-registration, before anybody knew.

Handing off

The result is on the table and the two guns are loaded and pointed at it, by the person holding the result, on purpose, in writing, weeks earlier.

Episode 18 fires them.

Questions

1. **Commit now**, before episode 18. Does the Gospel result survive deleting the `unan` half? Write yes or no and one sentence of reasoning.

2. Three of four, with a p of 0.00034. **What would you put on screen** if the series ended here?

3. Name a sensitivity analysis you could pre-register on your own current work **that could genuinely overturn it.** If you cannot think of one, that is the finding.

Collecting episode 17

Three of four. p equals 0.00034. The ordinal series, finally, on the right question, with a bigger sample and frozen tiers.

And two analyses declared in advance, by the person holding the result.

S1, and it does not survive

```
core only:   GOSPEL 0.402   TRAGEDY 0.503   MYTH 0.325

             ordinal BREAKS.  Tragedy is now above the Gospels.
             GOSPEL > MYTH    p = 0.293        <- D2 is gone
```

Take out the half that was flagged as suspect, and the Gospel result evaporates. p goes from **0.00034 to 0.293**, which is nearly a thousandfold, **and the ordering inverts.**

> **The concepts that state Girard's actual claim carry none of the effect. The contaminated half carries all of it.**

And notice what that means about the disclosure. It was not a formality. It was not defensive throat-clearing. **The thing it warned about is exactly what happened**, and it was written down before anyone could have known.

S2, and it takes the other one

The second analysis asked a different question. The myth tier was 71 texts, but **37 of them are epic**, and somebody could reasonably object that epic is not myth.

So: rerun with epic excluded, myth restricted to mythography alone.

```
MYTHOGRAPHY 0.674   EPIC 0.382   TRAGEDY 0.667

TRAGEDY > MYTHOGRAPHY    p = 0.388        <- D3 is gone
```

Mythography scores above tragedy.

So D3, tragedy above myth, only passed because thirty-seven epic texts were dragging the myth tier's median down. Remove them and **the myth-below-tragedy half of the ordinal series does not exist.**

> What exists is not myth below tragedy below Gospel. What exists is **epic below everything**, which is a fact about epic and not about Girard.

So what is the honest headline

Nominally three of four. Actually **D1 alone**, and nothing that survives its own pre-registered controls.

Both surviving predictions were killed by analyses the project wrote against itself.

The thing that makes this worse than a failure, and better

Look at where the effect actually lived.

It was not spread thinly across the lexicon. It was concentrated, entirely, in the half that had been named in advance as the half most likely to be an artifact of the author knowing his corpus. **The prediction inside the disclosure came true**, which is a more specific and more damning result than the round simply failing.

And it repeats. `core` **has never carried anything, on any measure, in any round**, which by the end of the project is four consecutive rounds finding the same thing. That is not four failures. **It is one finding, confirmed four times**, and it is about the instrument rather than about myth.

Why this is the strongest episode in the season

Sit on this, because it is the whole argument for the method.

A project trying to look good does not write S1 and S2. They exist for one reason: they were registered before anyone knew what they would do. If they had been thought of afterwards, **they would not have been thought of**, because by then there was a result worth protecting.

> **The value of a sensitivity analysis is entirely in when it was written.** One written after the result is a robustness check. One written before is a loaded gun pointed at your own foot, and this one went off.

The clinical anchor

A pre-specified sensitivity analysis that overturns the primary endpoint.

Episode 17 said most pre-specified analyses are chosen because they will probably agree. **This is the other kind**, and you know how rare it is.

A pre-specified analysis that could destroy your headline, written by the person whose headline it is, is close to nonexistent in the literature.

> So the refrain has a second edge here. **A sensitivity analysis chosen because it will probably agree is a check that cannot fail**, and it prints reassurance in the most respectable way available: pre-specified, documented, and worthless. And when one

does overturn a primary, the paper usually leads with the primary and puts the sensitivity in a supplementary table, where it is true, published, and read by nobody.

The character anchor

The boxer from episode 17, and the bell has gone.

He said which round he had to win in. He did not win it. **And the only reason anybody knows is that he said it first**, because a fight nobody set a condition on is a fight you can always describe as close.

What the round is still worth, and it is not nothing

Do not let the collapse erase what survived, because something did.

D1 held. The medians came out in the predicted order, on the right construct, on 98 texts, and no sensitivity analysis touched it. That is one clean pre-registered pass on the interesting half of Girard's claim, and it is the only one in the project.

And the instrument proved it can produce a positive. Three rounds had failed in ways that left an open question about whether the whole apparatus could ever return anything but null. **Round 4 answered that**, and then took most of it back itself.

> The honest summary is not "round 4 failed." It is **"round 4 produced one surviving ordering and demonstrated that its own effect lives in the half it had flagged as suspect."** Both halves of that sentence are results.

A round that discovers where its effect lives has learned something a round that merely passes never does.

What would count against this

1. **S1 being too strict.** Deleting `unan` removes the phrases most likely to express dissent in English, so the core-only run may be measuring a construct nobody could detect in any corpus. **A real objection**, and the counter is that the `core` concepts are the ones that state Girard's claim, so an instrument that cannot find them is not measuring his claim.
2. **S2 being the wrong exclusion.** Epic may genuinely belong in the myth tier, in which case D3 stands and this analysis answered a question nobody needed to ask. **The project does not adjudicate this**, and the honest position is that the tier boundary was always arguable.
3. **The whole round still resting on a rate.** Both analyses argue about which texts and which words, and **neither touches the fact that a whole-text rate discards position**, which episode 19 says is the actual cause of everything.

What to do with your face on camera

Do not present this as a defeat.

The round failed. The method worked. **Those are the same sentence**, and the whole series is about people who cannot hold both at once: the apologist who never loses an item, the skeptic who never loses one either, and the researcher whose robustness checks always agree with him.

Handing off

Two rounds have now failed on the interesting half of Girard's claim, for reasons that look completely different.

Episode 19 shows they are the same reason, and it was named in Season 2 before either round existed.

Questions

1. Nominally three of four, actually nothing. **Which is the honest headline**, and what would you put on screen?

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2. Would you have written S1, knowing it might do this? **Answer before you decide you would have.**

3. Mythography scored above tragedy and nobody predicted it. **What might explain it, and how would you test that without fishing?**

4. Go back to what you committed in episode 17. **Were you right?** If you were, what told you, and why did the project need the analysis to find out?

Collecting the season

Four rounds. One survived and it confirmed the half nobody disputed. This episode is what the other three have in common, and it is a single sentence that was available in Season 2 and took three failures to absorb.

The resolution

Round 3 failed because the statistic was invariant to word order. Shuffle the text and the number does not move, so it could not locate an episode. Episode 12.

Round 4 failed the same way, one level up, and this is the part worth the episode.

Girard's claim is not that Gospels contain more division-language than myths do. It is that **at the moment of collective violence**, a voice separates from the crowd.

And John's three `division` hits sit at chapter 7 verse 43, chapter 9 verse 16, and chapter 10 verse 19.

They are disputes about who Jesus is. They are chapters away from the crucifixion.

A whole-text rate counts all three and reports a high dissent score for John. There is no way for it to notice that none of them is anywhere near a killing, **because position in the narrative is exactly the information a rate discards.**

> **Two failed rounds, one cause: whole-text rates cannot test claims about narrative structure.**

That was planted in Season 2 episode 11, when the rate was introduced and its one property was named by shuffling a book into a bag. **It has now detonated twice**, first in episode 13 of this season and again here.

And the shape of the repetition is what makes it worth an episode rather than a line. Round 3 did not ignore the problem. **Round 3 was built specifically to fix it**, with a sliding window and a peak statistic and a shuffle null, all of which were the right idea. Then round 4 changed the construct from four stereotypes to dissent, kept the sharper question, and **quietly went back to a whole-text rate**, because a rate is what you reach for when you are counting a new thing.

So the lesson was learned about one instrument and did not travel to the next one. That is the ordinary way lessons fail, and it is much more common than forgetting them.

The character anchor

Looking for your keys under the streetlight.

Everybody knows the joke. You dropped them in the grass and you are searching on the pavement, because that is where you can see.

Now add the part that makes it this season. Somebody tells you the keys are in the grass. You accept it completely, you agree the light is the problem, and you go and buy **a brighter bulb.**

That is round 3 to round 4. The diagnosis was correct, the effort was real, the new equipment was genuinely better, and it was installed over the same patch of pavement. **A better instrument pointed at the wrong quantity is still pointed at the wrong quantity**, and the improvement is what makes it hard to see, because every measurement really did get sharper.

The clinical anchor

Ordering a more sensitive version of the wrong test.

The first assay came back unhelpful, so you order the high-sensitivity one. It is a genuinely better assay, it costs more, it detects smaller quantities, and the laboratory is right to offer it.

And it is measuring the same analyte, which was never the one that answers your question. The second result arrives with a tighter confidence interval and tells you nothing the first did not, **and it feels like progress because something improved.**

That is the failure this season keeps producing, and the reason it is hard is that refusing the better assay looks like refusing rigour. **The move that would have worked was to change what is being measured**, and nothing in the process of getting better at measuring ever suggests it.

The device: two rounds, one cause

	ROUND 3	ROUND 4
the construct	four stereotypes	dissent
the question	do they converge on a passage	is there a dissenting voice
the statistic	peak of a windowed minimum	**whole-text rate**
what the statistic discards	nothing, it was built for position	**position**
where the hits actually sit	the *Bacchae* peak at 54%, correctly	John 7:43, 9:16, 10:19, **chapters from the killing**
the failure	no localisation beyond chance	a rate cannot see the moment

> **Round 3 solved the position problem and round 4 threw the solution away**, because a rate is what you reach for when you start counting a new thing.

What the season produced, counted honestly

One clean positive result, at p equals 0.0045, confirming that persecution texts contain persecution vocabulary.

Fourteen failed pre-registered predictions.

Two failures for the *Bacchae*, on two different operationalizations.

Two null designs that could not have worked, one of which was invalid by construction.

And four occasions on which the project caught itself: refusing the unregistered metric in episode 1, refusing the free pass in episode 13, admitting the power problem was foreseeable in episode 15, and writing the sensitivity analyses in episode 18 that destroyed its own result.

Put that last count next to the first one and decide what you think, because the season does not decide it for you. **One clean pass against fourteen failures is a terrible record for a hypothesis and an excellent record for a procedure.** Every one of the fourteen was reported, and reported at the top rather than in a paragraph near the end, by the person who wanted them to go the other way.

The question the season leaves open, and it must be left open

Is Girard wrong, or is the instrument blind?

Nothing in this season settles it, and anybody who tells you otherwise is selling something.

The case for **blind**: every failure has an identified mechanism, and none of the mechanisms is about Girard. A minimum saturating on short texts is arithmetic. A rate discarding position is arithmetic. A null favouring even sets over clumpy ones is arithmetic. **None of those is evidence about myth.**

The case for **wrong**: at some point a theory that requires four instruments and never once produces its predicted effect is being protected rather than tested. And Girard's own flagship example has now failed twice.

And notice that the case for "blind" has a shape that should worry you. Every failure came with a mechanism, every mechanism was innocent of Girard, and each one arrived after the failure. That is exactly what a research programme looks like when it is being defended, and the fact that each individual explanation is correct does not settle it, **because a wrong theory investigated by a competent person also generates a correct explanation for every failure.**

The difference between the two situations is not visible in any single round. It is visible in whether the explanations ever stop being needed.

> Season 1 episode 10 asked you to write your answer down before you knew any of this. **Go and look at what you wrote.**

What comes next

Season 4 runs five more rounds, and two of them finally succeed at something.

And the thing they succeed at is smaller and stranger than anyone expected, and it belongs entirely to the half of the lexicon that was flagged as contaminated before it was ever scored.

What would count against this season's verdict

1. **The single cause being a frame rather than a finding.** Two rounds failed for reasons that look identical once you have decided to look for one reason. **Test it the other way: name what round 3 and round 4 would have had to do differently**, and if the answer is the same sentence for both, the cause is real.
2. **The instrument being blind for a fifth reason nobody has named.** Four are identified. **There is no argument anywhere that four is all of them**, and the project's record of finding a new one per round is not encouraging.

3. **Localization not being the bottleneck at all.** If the stereotypes are diffuse properties of whole texts, then every round measured the right thing and found nothing because there is nothing. **Episode 11 flagged this and nobody in the project entertains it**, which remains the most suspicious silence in the repository.

Questions

1. Answer it. Wrong, or blind? Write it down before Season 4.
2. What measurement would settle it? Describe it before you are shown one.
3. Fourteen failures and one uncontroversial pass. Is this a failed project? Be careful, because the answer defines what you think the series is about.

Runtime

Do not trust a runtime estimate taken from this document's length. Run `audit.py`. It reports the speakable word count per episode with tables, code and question blanks stripped, which is the only number that predicts anything. Divide by 135.

The toolbox introduced this season

TOOL	EPISODE	THE ONE-LINE VERSION
Poisson zero probability	2	rare things are absent from small samples, and absence is not evidence
scale validation	3	validate a metric at the length you will use it
clean test corpus	4	the texts that caused the change cannot test the change
translation control	5	if two translators disagree, no comparison means anything
dilution	6	a rate cannot see a focal event
density confound	7	a rate rewards plot per word
frequency-matched null	9	could random words this common have done this
what a null holds constant	**11**	**the most transferable idea in the series**
power	14	a negative from a blunt instrument is not a negative
sensitivity analysis	16	register one that could kill your result

Numbers to have cold

Aglauros, a declared control, MIN 1.27, beating every HIGH text	ep 1
Pentheus, MIN 0.00	ep 1
`crimes` 0.95 per 1000; P(zero) at 1,500 words = 0.24	ep 2
two *Agamemnon* translations 17.2% apart, n = 1	ep 5
Bacchae rank 35 of 72; 17,180 words, one messenger speech	ep 6
mythography 13.95 against tragedy 13.88; 28.65 event-words per 1000	ep 7
p = 0.0045, 8 of 2000	ep 9
round 3: z median 0.00, 4 of 71 against 3.6 expected	ep 12
`peak_min` median 2	ep 14
S1 takes p from 0.00034 to 0.293	ep 16
John's division hits at 7:43, 9:16, 10:19	ep 17

Progress log

DATE	EPISODE	WHAT WENT FUZZY ON CAMERA